Cavity-enhanced large-period lattice with graded-phase modes for 10^5 to 10^6 Cs sites

Technical report, working edition (generated from `docs/cavity_concept.md`)

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Technical report. Question (USER): for scaling a neutral-atom array with meta-optics to 10^5 to 10^6 sites, the limits are laser power, heat dissipation and damage by defects; can a combination of a build-up cavity and meta-optics cut the power requirement by an order of magnitude, and which design and simulations support it? All numbers are SIM (review before submission) or DERIVED unless marked; SOURCE values were retrieved and read in this

session. Code: `src/metatrap/cavity/`, `scripts/cavity_*.py`, `configs/cavity_*.yaml`; results in `results/cavity_*/` (one params.json per run with config, git state and versions). LaTeX editions: `docs/report/` (`cavity_report.tex`, edited; `cavity_concept_faithful.tex`, generated from this file).

Result

A build-up cavity cannot give the static meta-optic tweezer array a tenfold reduction in power. A cavity-enhanced lattice can give more than a hundredfold, provided the mode that fills a 4 mm array is shaped with graded-phase mirrors and made non-degenerate with an intracavity telescope. The binding constraints then move from laser power and metasurface heating to mirror figure, alignment stability and the cleanliness of every surface near the beam waist.

(i) Tweezers through a cavity are bounded at $G \leq 3.2$. Any element inside the cavity is passed twice per round trip, so the meta-hologram (array efficiency 0.83) caps the enhancement at $1 / (1 - 0.83^2) = 3.2$, and the hologram with a second meta-objective at 1.8. A cavity built of mirrors alone (concentric, NA 0.5) has a round-trip wavefront error of 0.02 to 0.03 waves rms at 100 µm field and 2 to 3 waves at 1 mm, which limits a usable finesse to 3 to 5 over a few hundred sites. Retro-reflection gives at most $(1 + 0.8)^2 = 3.2$ with the meta-objective.

(ii) A 4 µm-pitch lattice from crossing sheet beams needs much less power than tweezers, and a cavity multiplies it. Each lattice axis is a self-intersecting ring whose mode crosses the atom plane twice at $2\,\theta = 15.4$ deg. For 10^6 Cs sites at a 0.2 mK barrier per axis (1075 nm):

Route	Power at the optics input
Meta-hologram tweezers (NA 0.55, this repository)	1.40 kW
Lenslet-array tweezers (this repository)	2.50 kW
Lattice in free space (two in-plane axes and one vertical axis)	1.04 kW
Cavity lattice, mirrors in vacuum, over-coupled to G = 300	3.5 W
Cavity lattice, mirrors in vacuum, G = 1500	0.69 W
Cavity lattice through the present cell, AR 0.025 % / 0.1 % per surface (G 298 / 88)	3.5 / 12 W
Cavity lattice through Brewster-angle windows (G 570)	1.8 W

At equal radial trap frequency instead of equal depth (41 kHz, which needs a 0.86 mK lattice barrier), the in-vacuum cavity lattice needs 1.4 W at G = 1500 and 6.9 W through good AR windows.

(iii) The flat-top mode is the design problem. A Gaussian mode flat to 1 % over the array would need a 32 mm waist and seven times more power than a flat-top. A graded-phase mirror makes a super-Gaussian flat-top an exact eigenmode (circulating intensity flat to 0.15 % rms, barrier depth uniform to 0.23 % rms over 10^6 sites, with negligible design loss). However, a 5.5 mm flat mode in a 150 mm ring is nearly degenerate: the nearest transverse mode is 0.017 rad per round trip away, with no loss discrimination. A $\lambda/1000$ rms figure error on the graded mirror then scrambles the depth by 31 % rms, and a 0.3 µrad tilt by 6 %. A 0.2× demagnifying telescope in each half trip removes the degeneracy. In the 50 mm ring all competing modes then lose ≥ 1.1 % per round trip against 1.4×10^{-8} for the fundamental; in the adopted 70 mm ring the nearest competitor lies 0.216 rad away with 3.3×10^{-3} loss (Section 2). The same $\lambda/1000$ error then gives 2.7 % rms, and a 1 µrad

graded-mirror tilt 0.4 %. The tightest remaining tolerances are $\lambda/1000$ rms figure on the large telescope mirrors (2.4 %) and sub-μrad tilt stability of those mirrors (1 μrad: 3.7 %).

(iii-b) The residual can be corrected by re-shaping the two graded mirrors, because each axis's depth map is the product of the two crossing beams' profiles and each graded mirror shapes one of them. With independent $\lambda/1000$ figure errors or 1 μrad tilts on all ten mirrors, which give 1.2 to 4.8 % uncorrected, closed-loop feedback on depths measured with the atoms (six rounds, no mirror metrology, insensitive to 1 % per-site noise) restores 0.22 to 0.54 % in four of six runs; the other two end at 1.9 and 2.3 % (the latter with 0.5 mm-scale errors). It needs a graded-mirror change of only 0.8 to 3.2 nm rms. A 2D model with non-separable errors gives the same picture (2.1 % to 0.50 % with an x-only correction). The feedback fails when the uncorrected mode is far from the design (a 22 % start diverged). A camera on the transmitted beams does as well, but not better (0.48 and 2.0 % for $\lambda/1000$, 0.22 and 0.24 % for tilts), and diverges from the same 22 % start. Open-loop correction from measured mirror maps needs 0.1 to 0.3 nm rms metrology on every mirror, after coating and as mounted, which is beyond what LIGO- or EUV-class interferometry provides in situ; such metrology serves fabrication, not correction. The vertical-lattice ring, whose beam is 6.9 mm wide, is the most sensitive (3.6 to 36 % uncorrected, 0.08 % after a perfect-metrology redesign). Correction does not rescue the plain ring without a telescope. All corrections are static; drift must stay below these levels between corrections.

(iv) Heat is no longer a constraint; defects are. Coating absorption heats the mirrors by at most 0.55 K when clamped (IBS, 1 to 10 ppm; Section 5b). Every surface near the waist of the in-plane rings sees 4.5 to 48 kW/cm^2 at 0.2 mK, comparable to the test masses of gravitational-wave detectors. The largest tolerable fully absorbing defect falls to 0.14 to 1.5 μm on fused silica, against 92 μm for the 200 W metasurface of the present design, and to 3.4 to 37 μm on sapphire. The heat itself is small: at 10^6 sites 10 to 99 mW is absorbed inside the vacuum system (IBS, 1 to 10 ppm), the mirrors need conducting mounts but no water cooling, and the binding thermal requirement is temperature stability of the ring (about 10 mK for a metal ring, kelvins for ULE; Section 5b). The vertical ring, which carries 785 of the 989 W, is worse: as first folded in Figure 5 its mirrors see 14 to 170 kW/cm^2 (defects ≤ 0.04 μm on fused silica, 1 μm on sapphire), and its 4 mm strip mirrors clip the narrow vertical flat-top (1.2×10^{-2} per round trip). The adopted design adds a 4× vertical telescope: 32 to 55 kW/cm^2 (3.0 μm on sapphire), strips of ±2 mm (P1) and ±6 mm (P2, G), which fit the fold, and atom feedback to about 1 % from 7.7 and 13 % (Section 5).

(v) Meta-optics belong outside the cavity. A pixelated metasurface as the graded mirror is too lossy with present fabrication: 5 nm rms width noise is 0.10 to 0.15 rad of phase noise, which costs about 1 % per bounce and limits G to about 36. The graded mirrors should be continuous freeform surfaces under an IBS coating. The meta-optics carry the input mode converter (watts, not kilowatts), the imaging and addressing objective (unchanged), and the loading/rearrangement tweezers.

(vi) The arrangement fits around the atoms (Figures 3 and 5). The two in-plane rings lie in the atom plane, the vertical-lattice ring in a vertical plane on one diagonal, folded entirely below the atoms, and the space above the atoms stays free for the NA 0.5 imaging objective. A standard MOT is blocked, but a six-beam MOT with its axes tilted about 55 deg from vertical fits with ≥ 15.9 mm clearance for 20 mm beams, and four further lines remain for optical pumping and Raman beams (Section 6b). A single-beam grating MOT also fits geometrically, but demonstrated Cs grating MOTs (7×10^6 atoms) are one to two orders of magnitude short of a single-shot load of 10^6 sites.

(vii) Noise sets the next requirement (Section 6c). For a 10 s heating time of 10 μK atoms at 0.2 mK, the input needs a relative intensity noise below -106 dBc/Hz at 40 kHz, the residual

laser-cavity detuning must stay below 22 Hz/$\sqrt{\text{Hz}}$ (7 kHz rms, 1.7 % of the linewidth) at G = 300 and five times less at G = 1500, and each ring mirror must move by less than 0.32 pm/$\sqrt{\text{Hz}}$ at the 20 kHz trap frequency. The last calls for a monolithic ring assembly with controlled mechanical resonances.

(viii) The large step is from 10^5 to 10^6 sites (Section 6d). The same models at 10^4 and 10^5 show that a plain ring without telescope is non-degenerate there (nearest mode 0.78 and 0.17 rad away, with 2.4×10^{-2} and 2.5×10^{-4} loss), needs 0.23 and 0.50 W of input at G = 300 (tweezers: 14 and 140 W), gives 1.6 to 3.0 % for $\lambda/1000$ errors on every mirror (the atom loop corrects 10^5 to 0.20 to 0.49 %), and keeps the in-plane mirrors at 4.2 kW/cm^2 (defects up to 1.6 µm on fused silica). The 10^6 telescope design does not scale down (4.4 % ideal at 10^4). At 10^6 the degeneracy brings the telescope, a 0.14 µm defect limit, up to 55 kW/cm^2 on the vertical ring with its vertical telescope, a threefold tighter angle stability than at 10^5, and a 10^9-atom MOT.

Main limitations. One-dimensional Fox-Li model of the horizontal profile with a separable Gaussian vertical mode, checked by one 2D case per question (ideal design, non-separable errors and their correction); scalar optics; static perturbations, one or two random realisations per case; noise treated as requirements from analytic heating rates, without a time-dependent simulation or a vibration model of the mounts; the vertical ring as a product of two 1D modes; the layout is a first-order design.

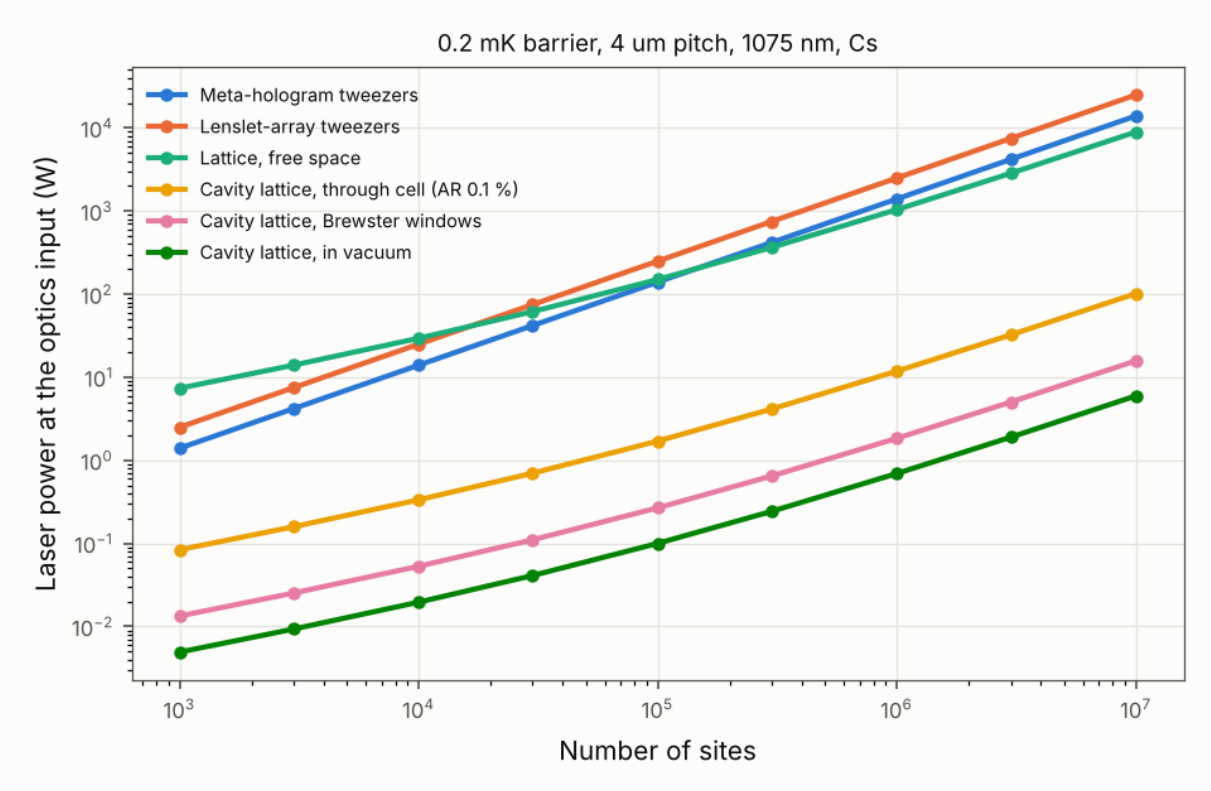


Figure 1. Laser power at the input of the trapping optics versus the number of sites, for a 0.2 mK barrier (tweezer depth) at 1075 nm and 4 µm pitch. Tweezer lines: 143 (hologram) and 80 (lenslet) mK x site / W from the Phase A simulations. Lattice lines: two in-plane axes and one vertical axis (8 µm period), flat-top x Gaussian sheets (w_z = 75 µm), mode matching 0.95; cavity enhancement for the over-coupled design T = 3L with 10 reflections of 50 ppm per round trip, plus 8 AR surfaces of 0.1 % (through cell) or 8 Brewster surfaces of 1×10^{-4}. Data: `results/cavity_budget/budget.json`.

Design summary

Parameters of the adopted design for 10^6 sites at a 0.2 mK barrier per axis (1075 nm, Cs). Sources in the text: Sections 2, 3, 4, 4b, 5 and 6c.

Quantity	Each in-plane ring (two)	Vertical ring
Lattice period, crossing angle 2 θ	4 µm, 15.4 deg	8 µm, 7.7 deg
Mode at the atoms	order-20 flat-top, 5.51 mm equivalent width, x Gaussian sheet w_z = 75 µm	flat-top 6.9 mm wide, x order-10 flat-top of 0.19 mm half-height
Telescope per half trip	0.2× Keplerian, f1 = 100 mm, P1 at 70 mm, L = 100 mm	0.15×, f1 = 160 mm, P1 at 110 mm, L = 120 mm
Round trip, free spectral range, reflections	1.16 m, 258 MHz, 10	1.66 m, 181 MHz, 10
Circulating power	102 W	785 W
Input power at G = 300 (mode matching 0.95)	0.36 W	2.75 W
Linewidth (FWHM) at G = 300 / 1500	413 / 82 kHz	289 / 58 kHz
Peak intensity on the mirrors	4.5 (P1), 48 (P2), 17.4 (G) kW/cm^2	32 to 55 kW/cm^2 with the adopted 4× vertical telescope (170 kW/cm^2 without it; Section 5)
Vertical telescope, strip half-heights in the ring plane	none	4× Keplerian (f_{1z} = 36.8 mm); ±2 mm on P1, P3, ±6 mm on the other mirrors; sapphire substrates
Trap frequency along the lattice axis	19.8 kHz	9.9 kHz
Depth uniformity: design / λ/1000 errors on all mirrors / after atom feedback	0.23 % / 2.0 to 4.0 % / 0.42 to 0.75 % (70 mm ring)	0.073 % / 5.4 to 12 % / 0.96 to 1.06 % (best round)
Static figure and tilt before correction	λ/1000 rms, < 1 µrad	λ/1000 rms
Noise for a 10 s heating time at 10 µK	RIN < -106 dBc/Hz at 40 kHz; 0.32 pm/$\sqrt{\text{Hz}}$ per mirror at 20 kHz	RIN < -100 dBc/Hz at 20 kHz; 0.64 pm/$\sqrt{\text{Hz}}$ per mirror at 10 kHz

Total circulating power 989 W; total input 3.5 W at G = 300 (0.69 W at G = 1500), against 1.04 kW for the same lattice in free space and 1.40 kW for meta-hologram tweezers. Input powers per ring are DERIVED from the circulating powers (`results/cavity_budget/budget.json`) as P_{circ} / (0.95 G).

1. Why the tweezer array cannot be cavity-enhanced tenfold

Enhancement. A ring cavity with input transmission T and all other round-trip losses L has G = T / (1 - r)^2 on resonance, r = sqrt((1 - T)(1 - L)), with the maximum G = 1 / L at T = L (`src/metatrap/cavity/budget.py`; `tests/test_cavity.py`). For G = 10 the round-trip loss must stay below 10 %. An NA 0.5 tweezer array is not an eigenmode of a mirror pair; the focusing optic must sit inside the cavity. Hence (`results/cavity_tweezer_bounds/bounds.json`):

Intracavity focusing element	Passes per round trip	$G_{\max}$
Meta-hologram, efficiency 0.83	2	3.2
Pair of meta-objectives, 0.80 each	4	1.7
Pair of refractive objectives, 0.95 each [VALUE?]	4	5.4
Ideal AR relay, 0.995 per pass [VALUE?]	4	50

Only the last line reaches 10, and it is not a meta-optic. A degenerate concentric cavity avoids intracavity lenses: two spherical mirrors share their centre at the atoms and image every focus onto itself. A ray trace of one reflection (NA 0.5, mirror radius 30 / 50 mm, piston, tilt and focus removed) gives a round-trip wavefront error of 0.031 / 0.019 waves rms at 100 μm field, 0.28 / 0.17 at 300 μm and 3.1 / 1.9 at 1 mm. Each site must stay within a fraction of the linewidth 2 π / F, so the finesse is limited to 3 to 5 at 100 μm and below 1 at 300 μm. The mirrors would also fill both NA 0.5 cones that imaging needs. Retro-reflecting the tweezer light through a second objective onto a mirror gives at most $(1 + \mathrm{sqrt}(\eta_{\mathrm{return}}))^2$, i.e. 3.2 for the meta-objective (η_{return} = 0.8^2 x 0.99) and 3.8 for a refractive one. It also forms a $\lambda/2$ axial standing wave in every trap. Hence the order-of-magnitude saving must come from a different trap geometry, one whose light does several jobs per photon (each lattice beam crosses every row of the array) and whose mode can be supported by mirrors alone.

2. Architecture

Three build-up cavities, one per lattice axis, mutually detuned by $\geq$ 100 MHz so that their interference averages out (0.36 ppm of the laser frequency; site shift 0.7 nm at the array edge).

In-plane axes. Each is a self-intersecting ring. Its mode crosses the atom plane twice, on arms A and B, at a full angle 2 θ with sin θ = λ / (2a) = 0.134 (θ = 7.72 deg for a = 4 μm at 1075 nm). Both crossing beams travel in the same direction, so the only fringe period is a; a standing-wave cavity would add $\lambda/2$ fringes along the beams. The two beams of one axis are the same circulating mode, so their relative phase needs no separate lock. The polarisation is vertical, perpendicular to the crossing plane, which gives full contrast. Horizontal polarisation would allow Brewster-angle windows at a contrast cos 2 θ = 0.964. The y-lattice ring has arms along x, and the x-lattice ring is the same ring rotated by 90 deg.

Mode. A sheet: flat-top in the plane, super-Gaussian amplitude exp(-|u/w|^{20}) with intensity $\geq$ 0.99 of the flat level over the array's transverse range |u| $\leq$ (S/2)(cos θ + sin θ) = 2.25 mm (S = 4.0 mm for 1000 $\times$ 1000 sites), giving w = 2.93 mm and an equivalent width of 5.51 mm. Vertically Gaussian with a 75 μm half-thickness at the crossing. Its Rayleigh range of 16.4 mm keeps the depth roll-off along the beams within 1 % over the array.

Graded-phase mirrors and telescope. Each half trip (crossing, arm A, to crossing, arm B) is symmetric about a graded-phase mirror G. With the target flat-top F at the crossing, the phase ϕ_G = -2 arg(K F), K the propagation from the crossing to G, returns conj(K F), and the symmetric return path then gives conj(F) = F. Hence F is an exact eigenmode of any lossless symmetric half trip (the graded-phase resonator of Belanger and Pare, Opt. Lett. 16, 1057 (1991), DOI 10.1364/OL.16.001057, abstract read in session). The design adopted in Section 4 puts a 0.2$\times$ Keplerian telescope of cylindrical mirrors in each half trip: primary P1 (f = 100 mm) 50 mm from the crossing, secondary P2 (f = 20 mm) 120 mm further, and G 100 mm beyond P2. Crossing to G 270 mm (half trip, crossing to crossing, 540 mm), round trip 1.08 m, free spectral range 278 MHz, 10 reflections per round trip. The layout of Figure 3 moves the primary to 70 mm from the crossing, because at 50 mm the mirrors of the two in-plane rings overlap by 4.2 mm: crossing to G 290 mm, round trip 1.16 m, free spectral range 258 MHz, angles of incidence 15 deg (P1), 44 deg (P2) and 24 deg (G). The 70 mm ring is equivalent in the Fox-Li model: depth uniform to 0.23 % rms, fundamental loss 1.6×10^{-8} per round trip, nearest competitor 3.3×10^{-3} (against 1.1×10^{-2} at 50 mm), and $\lambda/1000$ errors on all ten mirrors give 2.0 and 4.0 % uncorrected, 0.42 and 0.75 % after six rounds of atom feedback (`results/cavity_foxli`, `design_telK_M0.2_d70`; `results/cavity_correction_d70`). The other Fox-Li results of Sections 4 and 4b are for the 50 mm ring.

Vertical mode. For the 50 mm ring, a 30 mm vertical focal length on P1 and a 166 mm vertical radius on G give the 75 μm waist with a half-trip stability parameter m = 0.07, a transverse-mode

spacing of 3.0 rad per round trip (0.48 of a free spectral range), 0.24 mm on P1, 0.26 mm on P2 and 0.65 mm on G, and a 0.4 % waist change per 0.1 % radius error. For the 70 mm ring of the layout, 40 mm and 143 mm give m = -0.24, a mode spacing of 2.6 rad (0.42 of a free spectral range), 0.33 mm on P1, 0.18 mm on P2 (the vertical waist of 88 μm lies 81 mm behind P1, close to P2) and 0.55 mm on G, and 0.32 % per 0.1 % (`results/cavity_layout/layout.json`, both designs).

Vertical confinement. The in-plane sheets alone give 0.7 kHz vertically (Gaussian profile of 75 μm, 0.4 mK at the site). A third ring crosses at 2 θ_z in a vertical plane (period 8 μm, θ_z = 3.85 deg), propagating along the array diagonal. Its beams walk off vertically by ±0.19 mm across the array, so they must be flat-topped vertically as well (width 6.9 mm; equivalent height 0.57 mm with an order-10 edge, because an order-20 edge has a 4×10^{-4} design loss at this small size, `results/cavity_vertical`). This axis takes 785 W of the 989 W circulating at 10^6 sites and 0.2 mK; Section 3 discusses alternatives.

Vertical ring, fold geometry (Figure 5; `scripts/cavity_vertical_layout.py`, `configs/cavity_vertical_layout.yaml`; `results/cavity_vertical_layout/vertical_layout.json`). The ring is planar, in the vertical plane through the atoms that contains the array diagonal. A planar ring has no image or polarisation rotation, so the 6.9 mm flat-top axis and the polarisation stay horizontal, perpendicular to the ring plane (s-polarised at every mirror), and the cylindrical telescope mirrors curve perpendicular to their plane of incidence, where a parabolic cylinder is an exact cylindrical lens of focal length R / (2 cos AOI). Each half trip is symmetric about its graded mirror on the vertical axis. Both half trips fold below the atom plane: nothing of the ring rises above 9.4 mm, which leaves the space above the atoms to the objective. Because the crossing half-angle is small, the exit mirrors of the two half trips sit close together (P1 and P3, 14.8 mm apart vertically at 110 mm), and with an angle of incidence of at most 45 deg every fold must turn the beam by at least 90 deg. With the in-plane telescope (f = 100 mm, 0.2×) the best layouts found needed a 60 mm graded-mirror distance, which leaves a competing mode 0.07 rad away with 1×10^{-5} loss and gives 7.7 to 69 % for $\lambda/1000$ errors (exploratory Fox-Li check, not in the results files). A search over primary focal length, demagnification, distances and fold directions, with all clearances checked in 3D against the in-plane rings, selects: primary f = 160 mm, secondary f = 24 mm (0.15×), primary 110 mm from the atoms, G 120 mm behind the secondary, crossing to G 414 mm, round trip 1.66 m, free spectral range 181 MHz. Half trip 1 folds back under the atoms (its folded beam passes 30.6 mm from them) to G1 158 mm below the atoms; half trip 2 folds steeply down to 190 mm depth and back up to G2, 104 mm below the atoms. Clearances: 11.4 mm between mirror faces (strip mirrors 4 mm tall in the ring plane; Section 5 shows that P2, G and the return mirrors must be taller, which still fits), 11.1 mm between beams and foreign mirrors, 37.8 mm to the in-plane rings' mirrors and 35.8 mm to their beams; angles of incidence 8 to 45 deg. The effective propagation behind the telescope, L / M^2 = 5.3 m, is twice that of the in-plane rings (2.5 m), which partly compensates the wider beam (Section 4b).

Meta-optics. (i) Input mode converters, Gaussian to flat-top sheet (a Gaussian input couples 0.918 of the power into the flat-top mode; a flat-top converter is assumed to reach 0.95, `[VALUE?]`); (ii) the 852 nm imaging and addressing objective (`docs/metalens_report.md`, `docs/million_sites.md`); (iii) tweezers for loading and rearrangement. None of them sees the circulating power.

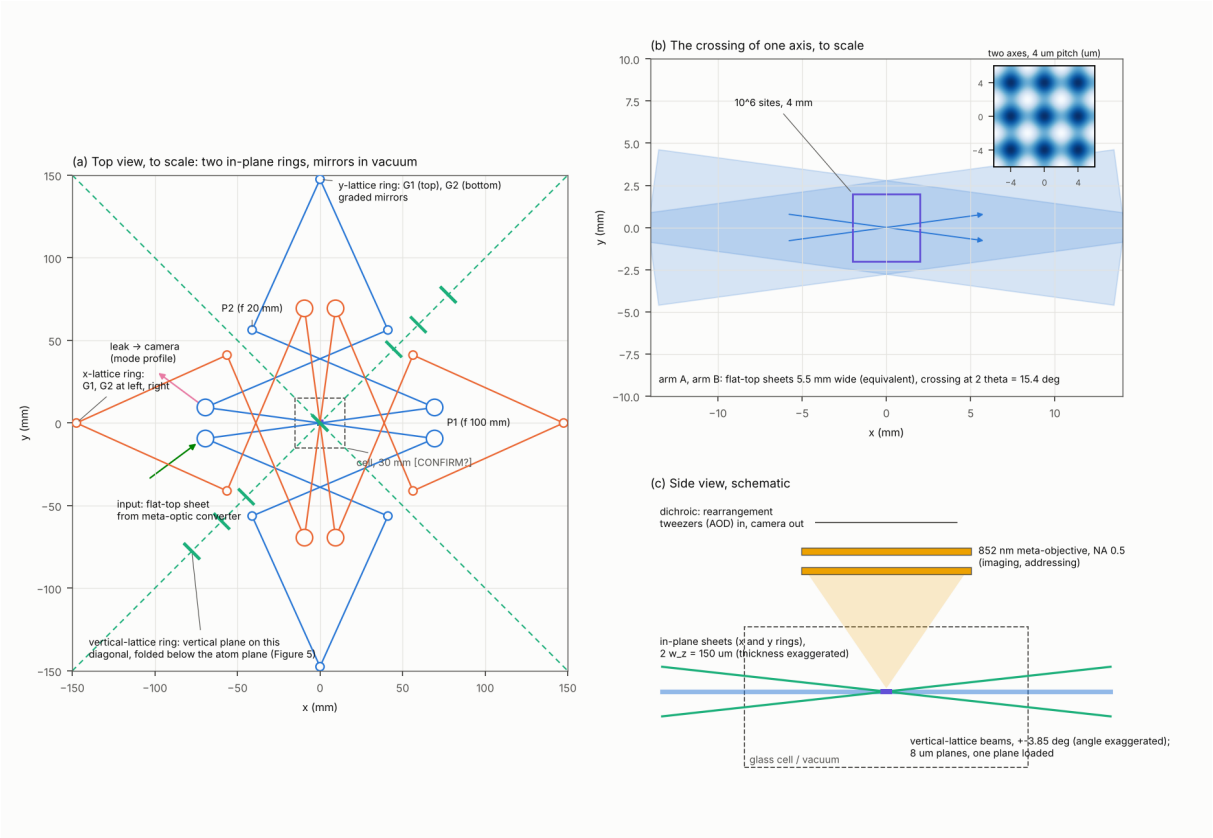


Figure 3. Arrangement. (a) Top view to scale: the y-lattice ring (blue) and the x-lattice ring (orange), each a self-intersecting ring with a 0.2× cylindrical telescope (primary P1, f = 100 mm, 70 mm from the crossing; secondary P2, f = 20 mm) and two graded mirrors G1, G2; circles are the mirror apertures (10 and 5 mm). Dashed diagonals: corridor for the vertical-lattice ring, whose fold geometry is shown in Figure 5 (the sketch predates it). The input (green) is the flat-top sheet from an extracavity meta-optic converter; the light leaked by a fold mirror (pink) goes to the mode camera. (b) The crossing of one axis to scale: the two flat-top sheets (equivalent width 5.5 mm, 2 θ = 15.4 deg) cover the 4 mm array; inset, the computed intensity of the two incoherent in-plane axes (plane waves, 4 µm pitch). (c) Side view, schematic, not to scale: in-plane sheets, the two vertical-lattice beams (±3.85 deg, 8 µm planes), and the 852 nm NA 0.5 meta-objective above the cell. Geometry from `results/cavity_layout/layout.json`; `scripts/plot_cavity_sketch.py`.

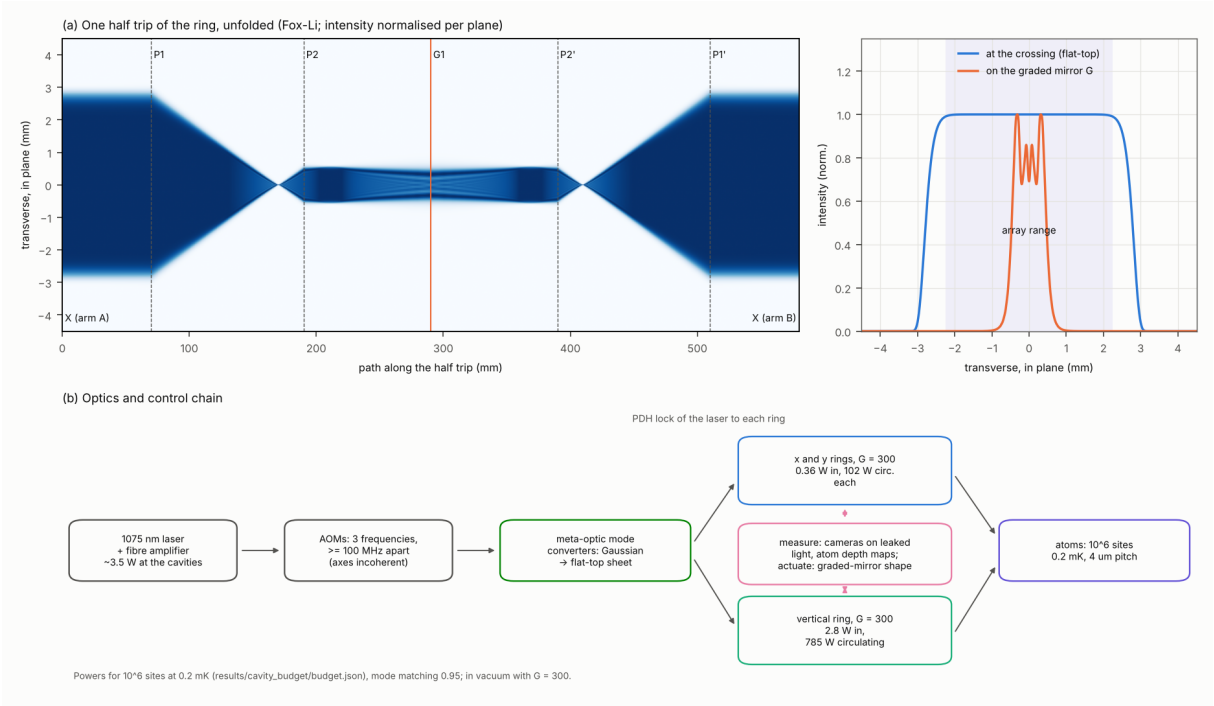


Figure 4. (a) The beam along one unfolded half trip of the 70 mm ring, from the crossing on arm A through P1, P2, G1, P2', P1' to the crossing on arm B (Fox-Li, design `telK_M0.2_d70`, 8192 points; intensity normalised in each plane). The flat-top is compressed fivefold by the telescope, passes a line focus between P1 and P2, and is shaped back by G1. Right: the profiles at the crossing and on G. (b) Optics and control chain for 10^6 sites at 0.2 mK with G = 300 (`results/cavity_budget/budget.json`, mode matching 0.95): 0.36 W into each in-plane ring (102 W circulating), 2.8 W into the vertical ring (785 W circulating). The graded-mirror correction of Section 4b closes the loop from the mode cameras and the atom depth maps.

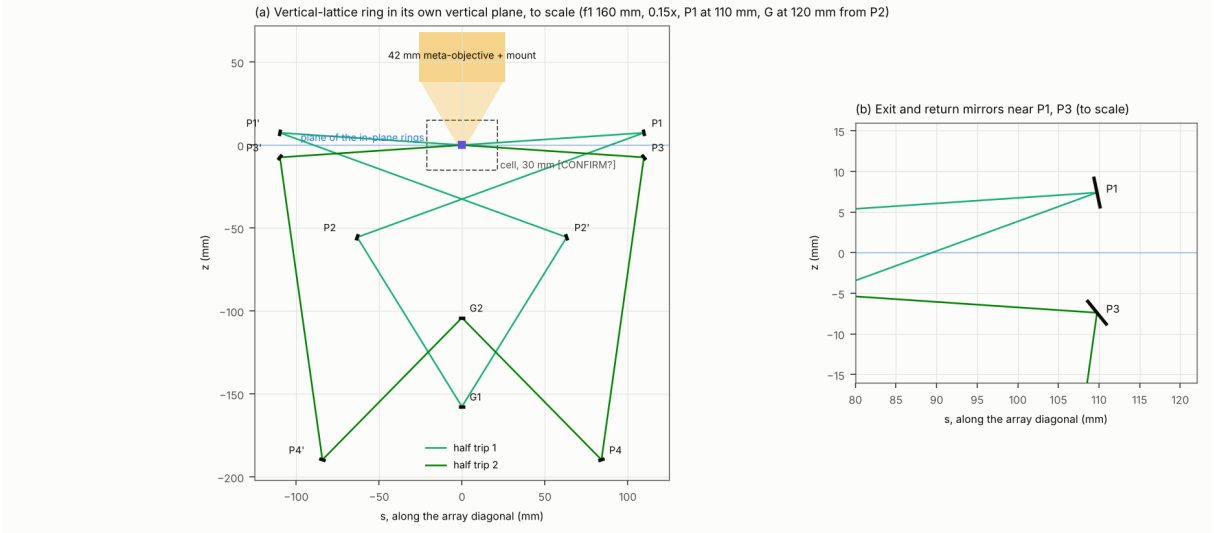


Figure 5. The vertical-lattice ring in its vertical plane through the array diagonal, to scale (s along the diagonal, z up). Half trip 1 (light green) leaves the crossing on the upper arm, folds at P1 back under the atoms to P2, the graded mirror G1 on the vertical axis 158 mm below the atoms, P2' and P1', and returns on the lower-left arm; half trip 2 (dark green) leaves on the lower arm and folds via P3, P4, G2 (104 mm below), P4', P3'. Black bars: mirror faces (4 mm strips in the ring plane as first laid out; the adopted design with a 4× vertical telescope needs ±2 mm on P1, P3 and ±6 mm on the other mirrors, Section 5; widths along the horizontal flat-top axis are 12 and 5 mm). Telescope f1 = 160 mm, f2 = 24 mm. Yellow:

the 42 mm imaging objective and its mount [CONFIRM?] and the NA 0.5 light cone. (b) The exit and return mirrors near P1, P3. Geometry from `results/cavity_vertical_layout`; drawn by `scripts/plot_cavity_sketch.py`.

Precedents. A two-cavity Sr lattice reached finesse 5025 and power enhancement 1132 at 914 nm, 489 µm waists, 80 mW per arm and ∼ 92 W circulating, in a monolithic ULE assembly in vacuum, with mirror losses ∼ 100 ppm. The authors report that the overlap sensitivity to misalignment scales as the fourth power of the waist, and attribute 1 % fringes to an accidental degeneracy with a high-order mode (Park et al., PRX Quantum 3, 030314 (2022), arXiv:2110.08073, full text read in session). Atom Computing used two self-intersecting cavities (finesse 2900 and 3000, waists 268 and 183 µm) for imaging 1225 Yb sites, with roughly two orders of magnitude lower power than tweezers and ≤ 50 ppm loss per mirror (Norcia et al., PRX Quantum 5, 030316 (2024), arXiv:2401.16177, HTML read in session). Both precedents cover arrays of 0.1 to 1 mm with Gaussian modes. Neither shaped a flat mode over several millimetres, and neither addressed the degeneracy that such a mode brings.

3. Power and loss budget

Model (`results/cavity_budget/budget.json`). Each crossing beam needs the peak intensity I1 = U / (4 c κ), with $\kappa = 2.53 \times 10^{-13}$ K per W/m^2 for Cs at 1075 nm (Steck polarisability, α = 1125 a.u.), i.e. 19.8 kW/cm^2 for U = 0.2 mK. The circulating power per in-plane axis is I1 x W_{eq} x sqrt(π/2) w_z, independent of the number of beams per axis because one mode supplies both. The in-plane power scales as the array side, sqrt(N), against N for tweezers; the vertical axis scales close to N, because its beams must also cover the vertical walk-off.

N, barrier	Per in-plane axis	Vertical axis	Total circulating	Hologram tweezers
10^5, 0.2 mK	32 W	78 W	143 W	140 W
10^5, 0.3 mK	48 W	117 W	214 W	210 W
10^6, 0.2 mK	102 W	785 W	989 W	1,399 W
10^6, 0.3 mK	154 W	1,177 W	1,484 W	2,098 W

The hologram column reproduces the 210 W of `docs/scaling_100k.md` for 10^5 sites at 0.3 mK. Round-trip loss and enhancement (10 reflections of 50 ppm; per-mirror bound of Norcia et al.):

Placement	Round-trip loss L	$G_{\max} = 1/L$	G at T = 3L (finesse)
Mirrors in vacuum	5.0×10^{-4}	2000	1500 (3140)
In vacuum, graded mirrors $+1 \times 10^{-3}$ each	2.5×10^{-3}	400	300 (627)
In vacuum, graded mirrors $+1 \times 10^{-2}$ each (metasurface)	2.0×10^{-2}	49	36 (75)
Through the cell, 8 AR surfaces at 2.5×10^{-4} [VALUE?]	2.5×10^{-3}	398	298 (623)
Through the cell, 8 AR surfaces at 1×10^{-3} [VALUE?]	8.5×10^{-3}	118	88 (183)
Through the cell, 8 Brewster surfaces at 1×10^{-4} [VALUE?]	1.3×10^{-3}	760	570 (1192)

We take the over-coupled coupler T = 3L as the design point. It gives 0.75 of the maximum enhancement at half the finesse, and the enhancement then depends weakly on the exact loss. Section 4 shows that the depth tolerances do not depend on the finesse between 47 and 7000. A

further reason to stay well below $G_{\max}$ is the cavity linewidth (82 kHz at G = 1500, 413 kHz at G = 300, for the 258 MHz free spectral range of the 70 mm ring; Section 6c). It converts laser frequency noise into intensity noise, which Norcia et al. identify as a heating source (190 ms time constant). We therefore recommend G $\sim$ 300 in vacuum (3.5 W for 10^6 sites at 0.2 mK) and use G = 1500 as the upper bound.

The vertical axis dominates at 10^6 sites because its beams must be flat over the vertical walk-off. Three options reduce it, none simulated here: (i) a larger vertical period (power scales as 1 / a_z; frequency as 1 / a_z); (ii) propagation along an array side instead of the diagonal (about half the power, at the cost of mirror clearance with one in-plane ring); (iii) a lower vertical barrier, if imaging tolerates it. Without the vertical axis the in-plane lattices need 205 W circulating at 10^6 sites, 0.7 W at G = 300.

4. The flat-top cavity mode (Fox-Li)

Method (`scripts/cavity_foxli.py`, `configs/cavity_foxli.yaml`; `results/cavity_foxli/foxli.json`). One-dimensional strip model of the horizontal profile on 8192 points over 24 mm (2.9 µm), exact scalar angular-spectrum propagation, hard apertures at every mirror, graded-phase design through the actual elements. The vertical profile is separable (a separable 2D propagation agrees with the product of 1D propagations to 2×10^{-3}, `tests/test_cavity.py`). The driven steady state (I - r e^{i ψ} R) E = t E_{in} is solved by GMRES. The laser is locked to the ψ of maximum circulating power, found by re-centring on the round-trip phase of the circulating field followed by a golden-section search. The field of arm A at the crossing and its image after the half trip (arm B) are propagated across the array, combined with the vertical Gaussian (width and Gouy phase), and sampled at 101 × 101 sites spanning the 10^6-site array. Metrics: rms and min / mean of the barrier depth, and the site displacement along the fringe normal with and without its affine part (a uniform pitch change or rotation is a calibration, not an error). Unless stated, the input is the target flat-top, G = 300, lumped loss 3×10^{-4}. Anchors: the strip Gaussian mode of a plane-concave design and its TEM1 Gouy phase (to 1×10^{-4}), and the driven enhancement against T / $(1 - r)^2$ (to 1×10^{-4}) (`tests/test_cavity.py`). The results do not change between 4096, 8192 and 16384 points (depth rms 0.23148 / 0.23149 / 0.23149 %).

Ideal designs. Every design reproduces the target. Over 10^6 sites the barrier depth is uniform to 0.23 % rms (min / mean 0.990, set by the 1 % design tolerance at the flat edge). The site displacement is 6 nm rms, all of it affine (residual 0.03 nm), from the differential Gouy phase of the two sheets.

Design	Fresnel number w^2 / (λ p)	Fundamental loss per round trip	Nearest competing mode (phase, loss)	Graded mirror PV, max deflection
Plane folds, p = 150 mm	53	8×10^{-10}	0.017 rad, 2×10^{-9}	5.9 rad, 0.004
Plane folds, p = 300 mm	27	1×10^{-9}	0.033 rad, 6×10^{-8}	9.0 rad, 0.004
Plane folds, p = 600 mm	13	3×10^{-6}	0.067 rad, 8×10^{-6}	13 rad, 0.003
Telescope 0.2×, Keplerian	-	1.4×10^{-8}	all modes lose $\geq$ 1.1×10^{-2}	24 rad, 0.013
Telescope 0.1×, Keplerian	-	2.3×10^{-8}	all modes lose $\geq$ 1.4×10^{-2}	36 rad, 0.026
Telescope 0.2×, Galilean	-	2.6×10^{-6}	0.33 rad, 1.5×10^{-3}	30 rad, 0.013

Spectra by full diagonalisation of the round trip on 4096 points. Edge order: n = 10 and 40 give the same uniformity (0.24 / 0.23 %) at equivalent widths of 6.78 / 4.98 mm against 5.51 mm for n = 20. In the telescope ring, n = 40 raises the design loss from 1.4×10^{-8} to 3.4×10^{-5}

and brings a competing mode to 4.7×10^{-3} loss, so n = 20 is the compromise between power efficiency and edge diffraction.

Why the plain design fails. With plane folds, the mode spacing is about λ p / w^2, the inverse Fresnel number of the flat mode: 0.017 rad at p = 150 mm and 0.067 rad at 600 mm, with no loss difference. A perturbation of round-trip phase $\delta\ \phi$ mixes the nearest mode into the fundamental with an amplitude of about $\delta\ \phi$ / 0.017. A $\lambda/1000$ surface error (12.6 mrad of reflected phase) is then an order-one admixture. This is the flat-top counterpart of the waist^4 alignment sensitivity of large Gaussian modes noted by Park et al. Curved folds do not help: a single cylindrical fold raised the spacing to at most 0.025 rad, and Fourier-transform configurations made it smaller (exploratory runs, not in the results file).

The telescope. A demagnifying telescope of magnification M makes the remaining path L to the graded mirror act, seen from the crossing, like a path L / M^2. The mode on G then has a Fresnel number near 1, the Belanger-Pare regime of strong mode discrimination. In the 0.2× design every competing mode loses at least 1.1 % per round trip, a million times the fundamental's loss.

Tolerances (median over two random realisations and correlation lengths of 0.5 and 2 mm referred to the crossing; figure errors as rms surface height over the beam, reflected phase 4 π h / λ; both graded mirrors perturbed independently; G tilts and curvature errors on one graded mirror):

Entries: barrier-depth non-uniformity over the 10^6-site array (rms, median over realisations) and, in parentheses, the lowest enhancement relative to the analytic value. K Keplerian, G Galilean; P1 rows: the same figure error or tilt on the incoming primary of both half trips, i.e. two of the four large telescope mirrors carry identical errors (Section 4b treats independent errors on all ten mirrors). Generated by `scripts/cavity_report_tables.py`.

Perturbation	Plane folds, 150 mm	Plane folds, 300 mm	Telescope 0.2× K	Telescope 0.1× K	Telescope 0.2× G
G figure, rms, 0.001 waves	31 % (0.61)	17 % (0.84)	2.7 % (0.97)	1.3 % (0.98)	6 % (0.94)
G figure, rms, 0.003 waves	49 % (0.42)	44 % (0.41)	7.6 % (0.78)	3.9 % (0.82)	18 % (0.59)
G figure, rms, 0.01 waves	59 % (0.15)	57 % (0.19)	21 % (0.36)	14 % (0.21)	24 % (0.08)
P1 figure, rms, 0.001 waves	-	-	2.4 % (0.99)	1.3 % (0.96)	8.5 % (0.90)
P1 figure, rms, 0.003 waves	-	-	7.4 % (0.90)	3.9 % (0.73)	25 % (0.41)
P1 figure, rms, 0.01 waves	-	-	26 % (0.14)	14 % (0.09)	60 % (0.08)
G x^2 sag at beam half-width, 0.001 waves	5.7 % (1.00)	2.8 % (1.00)	0.82 % (1.00)	0.3 % (1.00)	0.87 % (1.00)
G x^2 sag at beam half-width, 0.003 waves	18 % (0.98)	8.7 % (1.00)	2.6 % (0.98)	0.65 % (0.99)	2.9 % (0.99)
G x^2 sag at beam half-width, 0.01 waves	61 % (0.86)	31 % (0.96)	12 % (0.70)	2 % (0.90)	5.5 % (0.78)
G tilt, 0.3 µrad	5.7 % (0.92)	3.3 % (0.97)	0.26 % (1.00)	0.23 % (1.00)	0.27 % (1.00)
G tilt, 1 µrad	15 % (0.68)	11 % (0.83)	0.43 % (1.00)	0.25 % (1.00)	0.53 % (1.00)
G tilt, 3 µrad	67 % (0.39)	26 % (0.58)	1.1 % (1.00)	0.38 % (1.00)	1.4 % (1.00)
G tilt, 10 µrad	80 % (0.26)	73 % (0.33)	3.7 % (0.97)	1 % (0.99)	5.1 % (0.96)
P1 tilt, 0.3 µrad	-	-	1.1 % (1.00)	0.6 % (1.00)	0.73 % (1.00)
P1 tilt, 1 µrad	-	-	3.7 % (0.97)	1.9 % (0.99)	2.4 % (0.99)
P1 tilt, 3 µrad	-	-	11 % (0.83)	5.8 % (0.93)	8.1 % (0.91)
P1 tilt, 10 µrad	-	-	33 % (0.56)	20 % (0.61)	24 % (0.59)

Reading the table. (i) The telescope reduces the sensitivity to graded-mirror figure 6 to 11 fold and to graded-mirror tilt 20 to 60 fold. (ii) The large telescope mirrors now carry the tightest tolerances. A $\lambda/1000$ rms figure error on the two incoming primaries gives 2.4 % rms, and a 1 µrad tilt of both 3.7 %. (iii) The 0.1× telescope halves these numbers but doubles the intensity on the small mirrors (Section 5). (iv) The Galilean form is 2 to 3.5 times more sensitive to figure errors than the Keplerian at the same magnification, though slightly less sensitive to primary tilt, because its design loss and its nearest mode (0.33 rad, 1.5×10^{-3} loss) are closer to the fundamental. (v) The finesse barely matters. With a fixed $\lambda/300$ figure error, the depth non-uniformity is 5.2 to 5.7 % at G = 30, 100, 300, 1000 and 3000 (finesse 47 to 7159), because the admixture is set by the mode spacing, not by the linewidth.

Input mismatch is filtered by the cavity. A flat-top input 2 % too narrow or too wide, or offset by 100 µm, changes the depth uniformity by ≤ 0.022 percentage points and the enhancement by ≤ 0.6 %. A Gaussian input with the optimal waist couples 0.918 of its power and changes the uniformity from 0.23 to 0.42 % rms. Hence the extracavity mode converter, the meta-optic role, needs only modest accuracy.

A plain cavity without graded mirrors (plane folds and telescope, same flat-top input) gives 12 to 22 % rms and 0.86 to 0.87 of the enhancement: the flat-top is not its eigenmode.

Metasurface realisation of the graded phase. Quantising ϕ_G to 8, 16 or 32 levels gives 19, 12 and 4.5 % rms, at 0.14, 0.23 and 0.52 of the ideal enhancement. White pixel phase noise of 0.02, 0.05 and 0.1 rad gives 1.5, 3.8 and 7.4 % rms and reduces the enhancement to 0.89, 0.52 and 0.15 of the ideal. The repository's SiN libraries at 1075 nm (LPCVD and Si-rich, p = 0.45 to 0.5 µm, h = 1.3 to 1.6 µm, `data/library_main.h5`, `data/library_est_1075.h5`) give a phase slope of 19 to 31 rad/µm over the 2 π range. Hence 5 nm rms width noise is 0.10 to 0.15 rad in transmission, and about twice that in a reflective double pass (an estimate; no reflective library was simulated). The noise model sets white noise per 2.9 µm grid cell, so it overstates the mode distortion from 0.5 µm pixels, whose scatter leaves the cavity at large angles. The loss, about σ^2 per bounce, is model-independent. A metasurface graded mirror therefore needs sub-nanometre width control, or it must accept G ∼ 36. The required phase is smooth (24 rad PV, deflection ≤ 0.013), so a continuous freeform surface (diamond turning, magnetorheological or ion-beam figuring) under an IBS coating is the natural realisation `[REF?: freeform figuring at nm rms over mm apertures]`.

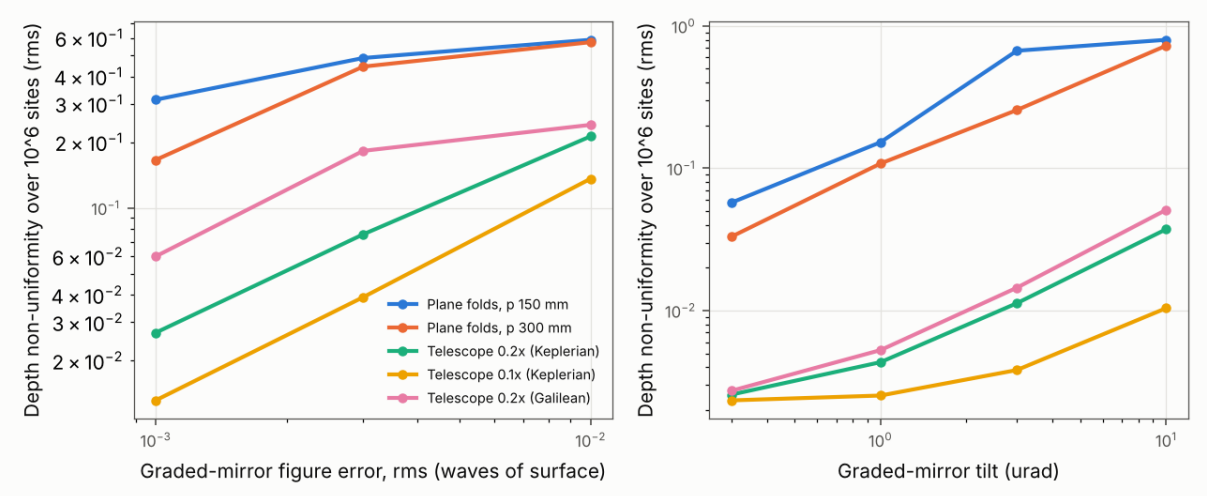


Figure 2. Barrier-depth non-uniformity over the 10^6-site array (rms over 101 × 101 sampled sites) versus the rms figure error of both graded mirrors (left, median over realisations) and the tilt of one graded mirror (right), for plane-fold rings of p = 150 and 300 mm and for telescope rings. Fox-Li strip model, 8192 points, G = 300, flat-top input, w_z = 75 µm. Data: `results/cavity_foxli/foxli.json`.

4b. Correcting the residual non-uniformity

Question. The ideal ring gives 0.23 % rms, but realistic mirrors give a few per cent (Section 4). A lattice offers no per-site trim. Its depth map per axis is the product $|E_A|$ $|E_B|$ of the two crossing beams' transverse profiles, and the two graded mirrors control them: G1 shapes the half trip from arm A to arm B, G2 the return. We ask how much of the error this pair can remove.

Method (`scripts/cavity_correction.py`, `configs/cavity_correction.yaml`; `results/cavity_correction/correction.json`). Independent smooth figure errors, random tilts, or both, on every mirror of the ring: 8 telescope reflections and 2 graded mirrors for the 0.2× ring, 4 folds and 2 graded mirrors for the plain 150 mm ring. Correlation lengths are referred to the crossing and scaled by the beam size on each mirror. Three corrections: (i) open loop: G1 and G2 redesigned through the half trips as measured, ϕ = arg(K_{out}ˆ+ T) - arg(K_{in} T), which reduces to the conjugation design for a symmetric, unperturbed half trip (`tests/test_cavity.py`). Metrology noise is a smooth random error map of the stated surface rms on every mirror, plus 0.1 µrad of tilt error when tilts are present; (ii) closed loop from the atoms, with no knowledge of the errors. Each round the barrier depth at 101 × 101 sampled sites is measured with 0 or 1 % multiplicative noise per site. It is fitted as a(ξ_A) b(ξ_B) by log-linear least squares with a smoothness penalty. The target profiles are multiplied by (a, b)$^{-0}$.5 and G1, G2 redesigned with the nominal model, for six rounds; (iii) the stroke, the rms and peak-to-valley surface change of G relative to the nominal design. Enhancement G = 300, the self-consistent lock of Section 4.

Telescope ring, 0.2× (depth rms over the 10^6-site array; two random realisations where shown):

Errors on every mirror	Uncorrected	Perfect metrology	0.1 nm metrology	0.3 nm metrology	1 nm metrology	Atom loop, 6 rounds	Camera loop, 0.5 % noise	+ 1 % pixel response	+ 1 % flat field
λ/1000 figure, 2 mm	2 / 4.8 %	0.49 / 0.54 %	0.68 / 0.61 %	1.3 / 1.4 %	3.9 / 4.6 %	0.48 / 1.9 %	0.48 / 2 %	0.5 / 2 %	0.59 / 2 %
λ/1000 figure, 0.5 mm	3.6 %	2.4 %	2.3 %	2.3 %	3.9 %	2.3 %	2.3 %	2.3 %	2.3 %
λ/300 figure, 2 mm	7.1 / 22 %	1.4 / 1.7 %	1.5 / 2 %	1.9 / 2.9 %	4.1 / 7.2 %	1.5 / 48 %	1.5 / 28 %	1.5 / 31 %	1.5 / 29 %
tilts, 1 µrad rms	1.2 / 1.8 %	0.43 / 0.58 %	0.53 / 0.48 %	1.3 / 1.3 %	3.9 / 4.2 %	0.22 / 0.23 %	0.22 / 0.24 %	0.26 / 0.26 %	0.42 / 0.39 %
λ/1000 + 0.3 µrad tilts	2.9 %	0.6 %	0.71 %	1.4 %	3.9 %	0.54 %	0.54 %	0.56 %	0.65 %

Plain ring, plane folds, p = 150 mm:

Errors on every mirror	Uncorrected	Perfect metrology	0.1 nm metrology	0.3 nm metrology	1 nm metrology	Atom loop, 6 rounds
λ/1000 figure, 2 mm	20 / 42 %	0.23 / 0.23 %	0.69 / 2.1 %	2.2 / 6.2 %	11 / 19 %	3.6 / 1.6 %
λ/1000 figure, 0.5 mm	24 %	0.26 %	0.7 %	2.2 %	11 %	0.94 %
λ/300 figure, 2 mm	42 / 1.3×10^2 %	0.24 / 0.25 %	0.73 / 2 %	2.2 / 6.2 %	11 / 19 %	49 / 37 %
tilts, 1 µrad rms	19 / 4.8 %	1.9 / 0.32 %	3.1 / 1.8 %	4.1 / 5.9 %	13 / 19 %	20 / 0.41 %
λ/1000 + 0.3 µrad tilts	56 %	1.9 %	3.1 %	4.1 %	13 %	44 %

What the tables show. (i) Perfect metrology removes 63 to 92 % of the non-uniformity of the telescope ring: $\lambda/1000$ errors on all ten mirrors fall from 2.0 to 4.8 % to 0.49 to 0.54 %, and 1 µrad tilts to 0.43 to 0.58 %. The floor arises because a phase error on a mirror away from G becomes partly an amplitude error at G and at the crossing, which a phase-only mirror cannot undo. In the test ring of `tests/test_cavity.py`, a redesign through a half trip with an error on its primary reduced the round-trip infidelity only fourfold, from 1.5×10^{-3} to 3.2×10^{-4}, not to zero (test output, not stored in a results file). Errors on a 0.5 mm scale leave 2.3 to 2.4 %. (ii) The open loop is only as good as the metrology. At 0.1 nm rms per mirror the gain is kept, 0.3 nm leaves 1.3 to 1.4 %, and 1 nm (3.9 to 4.6 %) is no better than no correction. Is 0.1 to 0.3 nm achievable? The best large-optics programmes reach it on the instrument. Advanced LIGO test-mass substrates were ion-beam figured to 0.08 to 0.23 nm rms over the central 160 mm (specification < 0.3 nm, power and astigmatism removed; LIGO Scientific Collaboration, Class. Quantum Grav. 32, 074001 (2015), DOI 10.1088/0264-9381/32/7/074001, arXiv:1411.4547, Table 4). The metrology interferometer was specified for a repeatability below 0.15 nm rms, with 0.1 nm as the goal (LIGO-E080508-v2). For EUV projection optics, mirrors were measured with 27 pm rms repeatability and 46 pm reproducibility, and an α-tool mirror reached 0.10 nm rms figure (Kurz et al., Carl Zeiss SMT, EUVL Symposium 2004). Three differences make our requirement harder than these. First, it is absolute, per mirror, and must hold after coating. Coating raised the LIGO end test mass from 0.18 to 0.69 nm rms and the input test mass from 0.15 to 0.31 nm (arXiv:1411.4547), more than our whole tolerance. Repeatability is not absolute accuracy: independent absolute calibrations of reference flats agree only to `[VALUE?: approx. 0.4 nm rms, from a search snippet, verify]`. Second, our mirrors are off-axis cylindrical or freeform surfaces used at 15 to 38 deg incidence, each needing its own null test. Third, the map must describe the mirror as mounted, in vacuum and at the circulating power, and must be registered to the beam to a fraction of the error correlation length (sub-0.1 mm on the small mirrors). Hence such metrology belongs to fabrication, to bring each mirror to $\lambda/1000$, and not to in-situ correction, which must be closed-loop. (iii) The closed loop needs no mirror metrology. Of its eight runs, six converge within four to five rounds to the perfect-metrology level or below: 0.48 % ($\lambda/1000$), 0.22 and 0.23 % (tilts, the design floor), 0.54 % (combined), 2.3 % (0.5 mm errors) and 1.5 % ($\lambda/300$, first realisation). One $\lambda/1000$ realisation stalls at 1.9 % with 0.94 of the enhancement. The $\lambda/300$ realisation that starts at 22 % (0.48 of the enhancement) diverges. Hence the feedback works when the uncorrected mode is close to the flat-top (starts up to ~ 7 % here) and fails when the mode is no longer the designed one. Per-site measurement noise of 1 % changes nothing, because the separable fit averages over 10^4 sampled sites. A lower loop gain does not help (keys `*_gain0.25_r12` and `*_gain0.1_r16`). At gain 0.25 for 12 rounds the stalled $\lambda/1000$ realisation ends at 1.89 % (1.93 % at gain 0.5), and the 0.5 mm case at 2.1 to 2.3 % with its enhancement falling to 0.85. From the 22 % start, gain 0.25 and 0.1 still collapse the enhancement to 0.02 and 0.07, although at gain 0.1 the depth rms falls to 12 %. Hence the stall and the divergence are properties of the update rule, which redesigns with the nominal model and a phase-only mirror, not of its gain. A model-based update, in which the measured depths are used to estimate the mirror errors, is the next candidate; it is not simulated here. (iii-b) Cameras instead of atoms. The light transmitted by a cavity mirror carries the circulating profile of each beam; a camera behind an imaging relay conjugate to the crossing (assumed ideal here) sees $|E_A|^2$ and $|E_B|^2$ directly. With 10 µm pixels, 0.5 % fresh noise per pixel and 48 bins, the same update rule converges exactly as well as the atom loop: 0.48 and 2.0 % for $\lambda/1000$ errors (atom loop 0.48 and 1.9 %), 0.22 and 0.24 % for tilts (0.22 and 0.23 %), and 2.3 % for 0.5 mm-scale errors. It also fails where the atom loop fails: from the 22 % $\lambda/300$ start it diverges (28 %, enhancement 0.03).[1] A 1 % static per-pixel response error changes the result by at most 0.06

[1] An earlier version of the camera model assigned each pixel to the lower bin edge, shifting the measured profile by half a bin. That version reported 1.3 % for the second $\lambda/1000$ realisation, convergence from the 22 % start (6.7 %), and a worse result for tilts (0.37 to 0.52 %). None of these differences survives the corrected binning

percentage points (the bins average it). A 1 % rms smooth flat-field error, which the loop cannot distinguish from a real profile error, adds at most 0.20 percentage points in the converged cases.

Separability (`scripts/cavity_2d_check.py`, `configs/cavity_2d_check.yaml`; `results/cavity_2d_check/check2d.json`). A 2D Fox-Li of the 0.2× ring (1024 × 192 points over 14 × 2.4 mm, vertical focal length 30 mm on the primaries, 2D conjugation design, G = 30 for speed). Isotropic smooth $\lambda/1000$ errors on all ten mirrors vary across the vertical beam on the small mirrors, so they are non-separable:

Case (2D)	Depth rms	Enhancement / analytic
Ideal 2D design	0.23 %	1.000
$\lambda/1000$ on all mirrors, uncorrected	2.1 %	0.999
Perfect metrology, full 2D phase on G	1.5 %	1.000
Perfect metrology, correction projected onto x only	0.53 %	0.999
Closed loop from the atoms, x-only updates, after 4 rounds	0.50 %	0.999

The ideal 2D design reproduces the 1D strip model (0.232 against 0.231 %), and the vertical mode on G (0.648 mm) reproduces the ABCD design of Section 2 (0.65 mm). Non-separable errors do not prevent the correction: an x-only (cylindrical) correction recovers most of it, and the closed loop reaches 0.50 % in four rounds, still falling. The full 2D phase-only redesign does worse than its own x projection. We attribute this to the phase-only matching of the whole 2D field, which spreads the residual amplitude error over the vertical profile, including the atom plane (not tested further).

Vertical ring (`scripts/cavity_vertical.py`, `configs/cavity_vertical.yaml`; `results/cavity_vertical/vertical.json`). The same 0.2× telescope in the horizontal direction, with a 6.9 mm equivalent width, and an order-10 flat-top of 0.19 mm flat half-height vertically. That is two 1D modes, each driven and locked separately. Ideal: depth uniform to 0.078 % rms, lattice planes flat to 9 pm peak-to-valley. $\lambda/1000$ errors on all ten mirrors in the horizontal direction give 3.6 % and 36 % for two realisations. The wider beam brings a competing mode to 0.14 rad with 3×10^{-4} loss. A perfect-metrology redesign restores 0.08 % / 0.085 %. In the vertical direction the same errors give 1.6 % / 2.8 %, corrected to 0.66 % / 0.61 %. The lattice planes stay flat to ≤ 0.25 µm peak-to-valley in every case, well inside the ±1.7 µm depth of focus of the imaging objective. A 0.1× telescope raises the competing mode to 0.28 rad (2.6×10^{-3} loss) but gives 11 % uncorrected for both realisations (exploratory run, not in the results file). Hence the vertical ring is the most error-sensitive of the three.

The adopted fold geometry (Section 2, Figure 5: 0.15×, f1 = 160 mm, P1 at 110 mm, L = 120 mm; `configs/cavity_vertical_chosen.yaml`, `results/cavity_vertical_chosen/vertical.json`) gives: in the wide direction a fundamental loss of 3.6×10^{-7} per round trip and a nearest competing mode at 0.30 rad with 7.8×10^{-3} loss (against 0.14 rad and 3.1×10^{-4} for the unconstrained 0.2× design); ideal depth uniform to 0.19 % rms (min / mean 0.988); $\lambda/1000$ errors on all ten mirrors in the horizontal direction 5.7 % and 12 % uncorrected (against 3.6 % and 36 % for the unconstrained 0.2× design), 0.19 and 0.20 % after a perfect-metrology redesign;

(`scripts/cavity_correction.py`, `camera_profile`). A 50 ppm mirror leaks ∼ 5 mW of a 100 W circulating beam (DERIVED), ample for a camera. Hence a camera loop can run continuously against drift, with the atoms as the occasional absolute calibration of the camera's flat field and of the relay. (iv) The stroke is 0.8 to 3.2 nm rms of graded-mirror surface after the atom loop (converged cases) and 1.1 to 8.5 nm for the perfect-metrology redesign. Hence the correction could be written by iterative refiguring, or applied in situ by a deformable or thermally actuated graded mirror. The loss and stability of an intracavity deformable mirror at kW/cm^2 is `[VALUE?]`. (v) The plain ring is fully correctable with perfect metrology (0.23 to 0.26 %), but not in practice: 0.1 nm of metrology leaves 0.7 to 2.1 %, and the closed loop converges in only four of eight error sets (from starts of 4.8 to 42 %); it fails in the others, one of them starting at 19 %. Correction does not replace the telescope.

in the vertical direction 1.5 and 0.7 %, 0.33 % after redesign. Atom feedback, with errors in both directions at once and the depth map fitted as a product of a function of the horizontal transverse coordinate (the product of the two beams' profiles) and a function of position along the diagonal (the vertical flat-tops at $\zeta = \mathrm{s}\,\sin(\theta_z)$), reaches 0.17 and 0.19 % in six rounds from 5.9 and 12 %. For this ring the fit is exact, so the loop converges to the design value. The narrow vertical flat-top is the weaker direction for power: its design loss rises to 2.9×10^{-5} on the longer path, and the vertical errors reduce its enhancement to 0.80 to 0.90 of the ideal even after correction, i.e. 10 to 25 % more input power for this ring. The lattice planes stay flat to $\leq$ 0.52 µm peak-to-valley.

With the 4× vertical telescope adopted in Section 5 (`configs/cavity_vertical_ztel4.yaml`, `results/cavity_vertical_ztel4/vertical.json`; a Keplerian telescope in the ring plane at the same mirror positions, P1 curved with $f_{1z} = 36.8$ mm), the ideal depth is uniform to 0.073 % rms and the vertical mode loses 1.2×10^{-8} per round trip. Its nearest mode lies only 0.118 rad away with 4.4×10^{-3} loss, and the lowest-loss competitor 1.54 rad away with 8.9×10^{-4} (converged between 4096 and 16384 points, `results/cavity_vertical_spectra_check/spectra.json`; the earlier values from coarse dense grids, 2.1×10^{-6} and 1.9 rad, were not converged). This nearby mode is the likely reason for the larger error sensitivity. $\lambda/1000$ errors on all ten mirrors give 5.7 and 12 % (horizontal) and 5.4 and 7.3 % (vertical) uncorrected; a perfect-metrology redesign restores 0.08 and 0.10 % horizontally but only 0.98 and 0.89 % vertically, the phase-only floor of this ring. Atom feedback with errors in both directions reaches its best value, 1.06 and 0.96 %, after five and six rounds (from 7.7 and 13 %), with the enhancement at 0.99 / 0.97 and 0.98 / 0.94 (horizontal / vertical mode) and the lattice planes flat to ≤ 0.3 µm peak-to-valley. Continued to 12 rounds, the same update rule drifts back to 2.3 to 2.7 % (`results/cavity_vertical_ztel4_r12`), so the loop must stop at its best round. An 8× telescope behaves the same way (0.98 and 1.06 % after six rounds, `results/cavity_vertical_ztel8`) but does not fit the layout (below).

5. Intensities, defects and heating inside the cavity

Model (`scripts/cavity_thermal.py`; `results/cavity_thermal/thermal.json`). Each beam's peak intensity at the atoms is I1, set by the barrier depth. On a mirror it is I1 times the one-dimensional peak ratio from Fox-Li (5.7 on P2 and 6.5 on G for the 0.2× design), times the vertical size ratio from Section 2 (70 mm ring of the layout). The defect criterion is that of `thermal/damage.py`: a fully absorbing disc of radius a on a half-space heats by I a / κ, limited to 50 K.

Surface (0.2 mK, 10^6 sites)	Peak intensity	Largest absorbing defect, fused silica / sapphire
Cell wall, 15 mm from the atoms [CONFIRM?] (through-cell variant)	14.6 kW/cm^2	0.47 / 11 µm
Telescope primary P1	4.5 kW/cm^2	1.5 / 37 µm
Telescope secondary P2	48.1 kW/cm^2	0.14 / 3.4 µm
Graded mirror G	17.4 kW/cm^2	0.40 / 9.5 µm
Vertical ring, mirror with 3× beam expansion (placeholder; superseded below)	2.2 kW/cm^2	3.1 / 75 µm

At equal radial frequency (0.86 mK) every intensity is 4.3× higher and the defect limits 4.3× smaller (0.033 to 0.35 µm on fused silica). For comparison, the present 18.4 mm metasurface at 200 W sees 75 W/cm^2 and tolerates 92 µm (`docs/prefab_checks_report.md`).

Vertical-ring mirrors (`scripts/cavity_vertical_mirrors.py`, `configs/cavity_vertical_mirrors.yaml`; `results/cavity_vertical_mirrors/mirrors.json`). The last row of the table assumed a mirror with a threefold beam expansion in both directions, which the adopted fold

geometry (Section 2, Figure 5) does not provide. Its 0.15× telescope compresses the 6.9 mm horizontal width on P2 and G, and the 0.19 mm vertical flat-top has no telescope, so it spreads only by diffraction. We compute the peak intensity of the beam cross-section on each mirror from the two 1D Fox-Li modes, with the 785 W that circulate in this ring at 0.2 mK, for the adopted design and for two variants that add a Keplerian telescope in the vertical (z) direction at the same mirror positions (magnification 4 or 8, $f_{1z} + f_{2z} = 184$ mm, i.e. P1 curved in the plane of incidence with $f_{1z} = 36.8$ or 20.4 mm):

Vertical ring, 0.2 mK	P1 / P2 / G / P2 return / P1 return	Smallest defect, fused silica / sapphire	Nearest z mode (phase, loss); fundamental loss (converged spectra)	λ/1000 z errors: uncorrected, perfect redesign (G / G0)
Adopted, no z telescope	32 / 170 / 145 / 112 / 14 kW/cm^2	0.041 / 0.97 μm (P2)	2.15 rad, 4.0×10^{-3}; 2.9×10^{-5}	1.5 and 0.69 % (0.58, 0.97); 0.33 and 0.33 % (0.93, 0.96)
z telescope 4×	32 / 51 / 55 / 55 / 35 kW/cm^2	0.13 / 3.0 μm (G)	0.118 rad, 4.4×10^{-3}; 1.2×10^{-8}	5.0 and 8.3 % (0.99, 0.98); 2.0 and 1.4 % (1.00, 1.00)
z telescope 8×	32 / 27 / 32 / 32 / 37 kW/cm^2	0.19 / 4.4 μm (P1 return)	0.042 rad, 9.7×10^{-3}; 1.6×10^{-5}	23 and 6.8 % (0.79, 0.98); 2.8 and 0.41 % (0.99, 1.00)

The ideal depth uniformity is 0.19, 0.073 and 0.16 % rms. Hence, as folded, the vertical ring puts up to 170 kW/cm^2 on P2, 3.5 times the in-plane maximum (48 kW/cm^2 on P2), and tolerates absorbing defects only up to 0.04 μm on fused silica. This corrects the earlier placeholder row. An 8× vertical telescope brings every mirror to 27 to 37 kW/cm^2, i.e. 1.4 to 1.9 times the intensity at the atoms, which cannot be lowered on P1 without moving it further from the atoms. The surface intensity is lower than the cross-section value by cos AOI (0.90 on P2 and 0.85 on G at the smallest angles of the layout). The vertical telescope also lowers the design loss of the narrow mode and keeps its enhancement at 0.99 to 1.00 after a perfect-metrology redesign (0.93 to 0.96 without it), but it brings a competing z mode close to the fundamental (0.118 rad for 4× and 0.042 rad for 8×, with 4.4×10^{-3} and 9.7×10^{-3} loss; spectra converged in `results/cavity_vertical_spectra_check`) and the uncorrected depth more sensitive to figure error (5 to 23 % against 0.7 to 1.5 %).

Mirror heights and layout (`scripts/cavity_vertical_beam_heights.py`, `results/cavity_vertical_beam_heights`; aperture losses: `scripts/cavity_vertical_apertures.py`, `results/cavity_vertical_apertures/apertures.json`; `scripts/cavity_vertical_layout.py` in check mode, `configs/cavity_vertical_layout_check.yaml`, `results/cavity_vertical_layout_check/check.json`). The fold layout of Section 2 assumed 4 mm strip mirrors in the ring plane, while the Fox-Li model above used ±5 mm apertures. The narrow vertical flat-top diffracts: its rms half-height grows from 0.20 mm on P1 to 0.66 mm on the returning P1', and its far wings set the loss. Without a vertical telescope the vertical mode loses 1.2×10^{-2} per round trip with ±2 mm strips, 2.1×10^{-3} with ±3 mm, 2.6×10^{-4} with ±4 mm and 4.7×10^{-5} with ±5 mm; the first is five times the whole loss budget at G = 300, so the ring as first laid out would not reach its enhancement. Enlarging only P2, G and the return mirrors suffices: ±2 mm on P1 and ±8 mm elsewhere gives 2.4×10^{-6}. With the 4× vertical telescope, ±2 mm on P1 and ±6 mm elsewhere gives 2.6×10^{-6}; the 8× telescope needs ±3 and ±12 mm (1.2×10^{-5}). Re-evaluated in 3D against the in-plane rings, the chosen fold keeps 10.8 mm (no vertical telescope, 2 / 8 mm strips) and 10.6 mm (4×, 2 / 6 mm) of clearance with nothing above 9.4 mm. Uniform ±5 mm strips and the 8× telescope both rise above the 10 mm limit under the objective (12.4 and 10.4 mm).

Adopted vertical ring. Two designs fit: no vertical telescope with taller strips (0.17 to 0.19

% after atom feedback, enhancement 0.80 to 0.90, 170 kW/cm^2 on P2, defects $\leq$ 0.97 µm on sapphire), and a 4× vertical telescope (about 1 % after feedback, enhancement 0.94 to 0.99, at most 55 kW/cm^2, defects $\leq$ 3.0 µm on sapphire). Our choice is the 4× telescope on sapphire substrates, for two reasons. First, the threefold lower intensity brings the vertical ring to parity with the in-plane rings (55 against 48 kW/cm^2 on their secondaries), and the enhancement stays within 6 % of the design. Second, about 1 % rms in the vertical barrier depth acts on the vertical confinement only `[VALUE?: tolerable vertical depth non-uniformity]`. If the vertical lattice must be uniform to better than 0.5 %, the untelescoped ring with taller strips is the alternative. At equal radial frequency the vertical ring circulates 1059 W instead of 785 W, and every entry scales by 1.35. The cavity therefore trades a kilowatt laser and a hot metasurface for watt-level input and mirrors at kW/cm^2. Point absorbers at these intensities limit the stored power of Advanced LIGO (Brooks et al., Appl. Opt. 60, 4047 (2021), DOI 10.1364/AO.419689, abstract read in session). The fully absorbing disc model is pessimistic for sub-micron particles, whose absorption cross-section is below the geometric one.

Thermal deformation of the graded mirror (102 W circulating, equal-area disc of 0.43 mm radius on a 1 inch, 6.35 mm fused-silica or sapphire mirror, surface bulge $(1 + \nu)\ \alpha$ int T dz):

Absorber on G	Absorbed power	Surface temperature rise	Bulge, centre to edge	Bulge rms over the spot
IBS HR, 1 ppm `[VALUE?]`	0.1 mW	0.05 K	0.004 nm	1×10^{-6} waves
IBS HR, 10 ppm	1 mW	0.5 K	0.035 nm	1×10^{-5} waves
LPCVD SiN metasurface (3.5×10^{-6})	0.36 mW	0.19 K	0.012 nm	3×10^{-6} waves
Si-rich SiN metasurface (5.9×10^{-4}), fused silica	60 mW	32 K	2.1 nm	5.6×10^{-4} waves
Si-rich SiN metasurface (5.9×10^{-4}), sapphire	60 mW	1.3 K	0.9 nm	2.4×10^{-4} waves

With IBS coatings the deformation is three orders of magnitude below the $\lambda/1000$ figure tolerance. A Si-rich SiN film inside the cavity would reach the tolerance on its own. Fed into the Fox-Li ring (`scripts/cavity_thermal_foxli.py`, `configs/cavity_thermal_foxli.yaml`; `results/cavity_thermal_foxli/thermal_foxli.json`), with a bulge profile proportional to the intensity on G (sharper than the thermal profile, hence conservative) on both graded mirrors of the 70 mm ring, the depth rms changes from 0.231 to 0.232 % for 10 ppm IBS and to 0.252 % for 100 ppm. The Si-rich SiN film raises it to 0.53 % on fused silica and 0.32 % on sapphire; the enhancement changes by $< 4 \times 10^{-4}$. A static bulge is also within reach of the closed loop of Section 4b. The vertical ring's mirrors absorb 7.7× more (785 W) on smaller spots; on its sapphire graded mirror the bulge is 0.12 nm for 10 ppm (Section 5b). The cell windows (through-cell variant) absorb 4 ppm per pass in bulk (0.4 mW at 102 W), which is negligible thermally. Their surfaces carry the defect constraint of the table above, the reason to prefer mirrors in vacuum.

5b. Heat load and cooling: are chillers needed?

Model (`scripts/cavity_heat_budget.py`, `configs/cavity_heat_budget.yaml`; `results/cavity_heat_budget/heat.json`). In steady state every watt that enters a ring leaves it in one of three ways (DERIVED): with the over-coupled coupler T = 3L and mode matching η

= 0.95, a fraction 1 - 0.75 η = 0.29 of the input is reflected to an external beam dump, and 0.75 η = 0.71 is dissipated in the ring as P_{circ} L. The loss L (10 reflections of 50 ppm, plus 1×10^{-3} on each graded mirror at G = 300) splits into coating absorption (1 or 10 ppm per reflection [VALUE?]), a fraction of the graded-mirror loss that is absorbed rather than scattered (0 for figured IBS mirrors, 1 as a worst case), and light scattered or transmitted out of the mode. Mirror temperatures follow from the axisymmetric conduction model of Section 5 with the faces cooled only by radiation in vacuum ($h = 4\,\epsilon\,\sigma\,T^3 = 5.2$ W/m^2/K for $\epsilon = 0.9$ [VALUE?]), and the rim either clamped to a mount at chamber temperature or radiating (a free mirror). The hottest mirror of each ring is its graded mirror (smallest spot): a 1 inch fused-silica mirror in the in-plane rings, a 12 × 5 mm sapphire strip (as an equal-area disc, 6 mm thick [CONFIRM?]) in the vertical ring.

10^6 sites, 0.2 mK, G = 300	Total, three rings	In-plane ring (each)	Vertical ring
Laser input	3.47 W	0.36 W	2.75 W
Reflected to the external dump	1.00 W	0.10 W	0.79 W
Absorbed in vacuum, IBS 1 / 10 ppm	9.9 / 99 mW	1.0 / 10 mW	7.8 / 79 mW
Scattered or transmitted out of the mode (to walls, mounts, cameras)	2.4 W	0.25 W	1.9 W
Hottest mirror, absorbed (10 ppm)		1.0 mW	7.9 mW
Its temperature rise: clamped / free (10 ppm)		0.55 / 0.66 K	0.07 / 5.3 K
Its surface bulge, clamped (10 ppm)		0.036 nm	0.12 nm
Worst case, graded-mirror loss fully absorbed: absorbed; clamped rise; bulge	2.1 W in total	0.10 W; 55 K; 3.6 nm	0.79 W; 7.4 K; 12 nm

At 10^5 and 10^4 sites the input is 0.50 and 0.23 W and the absorbed power 14 and 6.5 mW (10 ppm). At G = 1500 the input falls to 0.69 W at 10^6, while the absorbed power, set by the circulating power, is unchanged.

What the budget shows. (i) The heat to be removed is small. With IBS coatings, 10 to 99 mW is absorbed inside the vacuum system at 10^6 sites, and 2.4 W of scattered light is spread over the chamber walls and mounts. The reflected watt goes to an air-cooled dump outside. None of this needs water cooling: it is three orders of magnitude below the 1.4 kW that the tweezer route would pass through its meta-optics. (ii) The mirrors must be heat-sunk by conduction, not by water. A free sapphire strip in the vertical ring would warm by 5.3 K, with a radiative time constant of about 12 min; clamped in a conducting mount it warms by 0.07 K. A mount conductance of order 0.1 W/K keeps the whole mirror within 0.1 K of the mount at 7.9 mW (DERIVED). (iii) The graded-mirror loss must be scatter, not absorption. If the 1×10^{-3} per graded mirror assumed at G = 300 were absorbed, the vertical graded mirror would take 0.79 W and bulge by 12 nm, eleven times the $\lambda/1000$ figure tolerance; the in-plane one would bulge by 3.6 nm and warm by 55 K. The bulge reaches $\lambda/1000$ at an absorbed fraction of about 9×10^{-5} of the circulating power (vertical) and 3×10^{-4} (in-plane), i.e. ten times the coating absorption. This excludes absorbing metasurfaces as graded mirrors (Si-rich SiN, 5.9×10^{-4}) on thermal grounds as well as loss, and it is met by figured surfaces under IBS coatings [VALUE?: absorption of a figured graded mirror].

The real requirement is temperature stability. A uniform temperature change of the ring material is benign for the lattice: all paths scale together, the lock retunes the laser, and the angles do

not change. It shifts the resonance by $\nu\ \alpha$ per kelvin, which the lock must follow and which changes the 100 MHz detuning between rings if they drift differently. A temperature gradient through the base plate bends it and tilts the mirrors. For a 25 mm plate over a 150 mm mirror spacing [CONFIRM?] and the 1 µrad tilt tolerance of Section 4 (DERIVED, expansion coefficients [VALUE?: approximate, verify]):

Ring material (expansion coefficient)	Resonance shift per K	Through-plate gradient for 1 µrad tilt	Uniform change that shifts the resonance by 100 MHz
ULE glass (approx. 3×10^{-8} /K)	8.4 MHz	5.6 K	12 K
Invar 36 (approx. 1.3×10^{-6} /K)	0.36 GHz	0.13 K	0.28 K
Stainless steel (approx. 1.7×10^{-5} /K)	4.7 GHz	10 mK	21 mK
Aluminium (approx. 2.3×10^{-5} /K)	6.4 GHz	7 mK	16 mK

Hence a metal ring needs gradients held to about 10 mK, while a monolithic low-expansion assembly (as in Park et al.) tolerates kelvin-scale changes. Water cooling of the ring would work against both requirements. Flowing water is a source of vibration, and Section 6c requires each mirror to move by less than 0.32 pm/$\sqrt{\text{Hz}}$ at the 20 kHz trap frequency [VALUE?: vibration spectrum of a water-cooled mount]. A chiller loop is also a strong temperature disturbance at the plate unless it is itself regulated to millikelvins. Our recommendation is therefore: (i) no water cooling on the cavity assembly; (ii) mirrors clamped in conducting mounts on a monolithic ULE or Invar ring; (iii) the ring on the chamber or an optical table, with slow temperature control at the 0.1 K level (ULE) or 10 mK level (metal), passive or by thermoelectric control far from the mirrors; (iv) constant circulating power, so that the 12 min mirror time constant never produces a transient during an experiment. Chillers remain needed elsewhere: for the laser only on the tweezer route (kilowatt amplifiers), and very likely for the MOT coils of a 10^8 to 10^9-atom MOT, which are not designed here [VALUE?: coil dissipation]. The cavity lattice's own laser (3.5 W at 10^6 sites) is an amplifier class that is often air-cooled [VALUE?: verify for the chosen amplifier].

6. Site quality and registration

Trap frequencies (`scripts/cavity_lattice.py`; `results/cavity_lattice/lattice.json`; plane-wave model, numerical Hessian, equal to the analytic (π / a) sqrt(2U / m) to 1×10^{-6}):

Barrier per axis	In-plane	Vertical (8 µm)	Tweezer of the same depth, radial / axial
0.2 mK	19.8 kHz	9.9 kHz	41.1 / 11.5 kHz
0.3 mK	24.2 kHz	12.1 kHz	50.4 / 14.1 kHz
0.86 mK	41.0 kHz	20.5 kHz	85.3 / 23.8 kHz

At equal depth the lattice site has 0.48 of the tweezer's radial frequency, because its $\cos^2$ well spans the full pitch. Equal radial frequency therefore costs a factor 4.3 in depth and power. A cost that the power table hides is the light shift at the site: each of the three incoherent axes reaches 4 I1 there, twice its mean, so the atoms sit at 3U of light shift (0.6 mK for a 0.2 mK barrier), against U in a tweezer. Differential light shifts and their noise are three times larger at equal barrier depth.

Registration. A differential wavefront error dw between the two beams of an axis moves a site by dw / (2 sin θ) = 3.7 nm per nm. A crossing-angle change of 1 µrad changes the pitch by cot(θ)

x 1 μrad = 7.4 ppm and moves the edge sites by 15 nm. A wavelength change of 1 ppm moves them by 2 nm. A change dL of the optical path between the two crossings of one ring shifts that lattice rigidly by dL / (2 sin θ), 3.7 nm per nm; the shift is common to all sites and can be tracked, but it sets the path-stability requirement between the crossings. In the Fox-Li tolerance cases the non-affine site displacement is at most 37 nm rms for every telescope case with depth non-uniformity below 5 %. Hence site positions are stable to a small fraction of the 4 μm pitch, provided the crossing angle is held to ~ 1 μrad. The rearrangement and addressing optics must be registered to the lattice, as in the Atom Computing scheme.

Layout (Figure 3; `results/cavity_layout/layout.json`, `scripts/cavity_layout.py`). With the primary 70 mm from the crossing and the same fold angle (157 deg) for both rings, the two in-plane rings fit within ±150 mm. Their mirrors and beams keep at least 15.5 mm from each other (mirror apertures and a 3 mm beam allowance included), the folded beams stay at least 35.8 mm from the atoms, and both array diagonals stay clear by 8.2 mm beyond 40 mm radius, the corridor left for the vertical ring. The first sketch, with the primary at 50 mm, overlapped by 4.2 mm. The vertical ring (Section 2, Figure 5) occupies the vertical plane on one diagonal, below the atom plane, and keeps 37.8 mm from the in-plane rings' mirrors. Mounts, coils and the vacuum envelope are not designed.

6b. Optical access, MOT and loading

Model (`scripts/cavity_access.py`, `configs/cavity_access.yaml`; `results/cavity_access/access.json`, access.png). For every line through the atoms we compute the clearance of a beam of given radius (250 mm each way) from all twenty mirrors of the three rings, each with a 5 mm mount allowance, and from the imaging objective above the atoms. The objective is the 42 mm aplanatic doublet that images 10^6 sites (`docs/million_sites.md`), with its first plate 38 mm above the atoms and a 26 mm radius including the mount `[CONFIRM?]`. Ring beams are light only and may be crossed; windows are assumed wherever needed `[CONFIRM?: chamber]`.

Beam set	Result
Standard MOT, axes x, y, z (20 mm beams)	blocked: horizontal beams hit in-plane mirrors (-17.5 mm), the vertical beam hits the objective (-36 mm)
Standard MOT on the diagonals + z	blocked (-14, -6.8, -36 mm)
Orthogonal MOT triad, axes tilted 54 to 55 deg from vertical (azimuths 15, 135, 255 deg)	free, clearance $\geq$ 15.9 mm for 20 mm beams
Lines free for 20 mm / 10 mm beams	34 % / 45 % of all lines through the atoms
Four further lines for optical pumping and Raman beams (56.5 deg from vertical; azimuths 117, 155, 297, 333 deg), $\geq$ 15 deg from the MOT axes	clearance 26.9 mm for 10 mm beams, 21.9 mm even for 20 mm beams
Imaging and addressing	the objective's own port above the atoms (NA 0.5; the free band starts at 43 deg from vertical)

The free band of directions lies between about 43 and 75 deg from vertical at all azimuths, except near the vertical ring's plane: the objective closes the cone above, and the ring mirrors the horizontal. A six-beam MOT with its three axes along the body diagonals of a cube standing on a corner (each axis about 55 deg from vertical) fits with margin, and leaves room for pumping and Raman beams. Global qubit rotations can also use microwaves (9.2 GHz, no optical access), with single-site addressing through the objective.

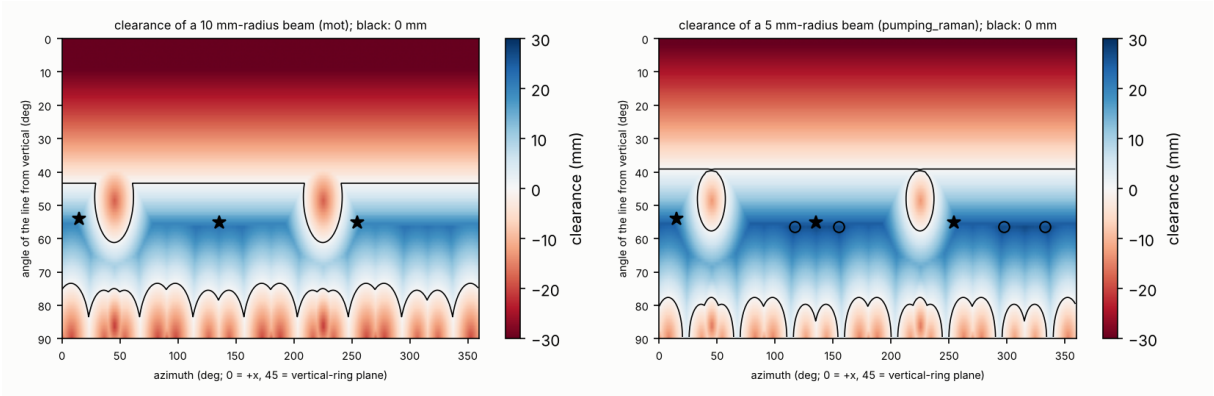


Figure 6. Clearance of a beam through the atoms versus its direction (angle of the line from the vertical, azimuth from +x; the vertical ring's plane is at 45 and 225 deg), for 20 mm (left) and 10 mm (right) beams; black contour: zero clearance. Each line through the atoms appears once (upper hemisphere). Stars: the three axes of the tilted six-beam MOT; circles: the four pumping / Raman lines. Obstacles: all ring mirrors with a 5 mm mount allowance and the 42 mm objective from 38 mm above the atoms [CONFIRM?]. `scripts/cavity_access.py`.

How many atoms does loading need (DERIVED)? A lattice site is $4 \times 4 \times 8$ μm^3 $= 1.3 \times 10^{-10}$ cm^3. With the usual collisional loading of about one atom in two sites [VALUE?: ~0.5 filling, light-assisted collisions], each site of the loaded plane must first see about two atoms, i.e. a density of $\sim 1.6 \times 10^{10}$ cm^{-3} across the 4 mm array. A MOT cloud that keeps this density at the array corners (2.83 mm) with a peak three to five times higher holds $\sim 5 \times 10^9$ atoms, because most of its atoms lie in other planes. Accumulating atoms in the in-plane sheets first (the sheets span ~ 19 of the 8 μm planes) reduces the need to $\sim 4 \times 10^7$ atoms delivered into the sheets, i.e. a MOT of 10^8 to 10^9 atoms of at least 6 mm extent. Demonstrated Cs numbers: 1.1(1)e9 atoms captured in an optical molasses from a $\sim 10^{10}$ atoms/s source (Donley, Heavner and Jefferts, NIST, `https://www.nist.gov/node/735766`, read in session); 2D-MOT fluxes of 4.02×10^{10} atoms/s (Huang et al., Chin. Phys. B 25, 063701 (2016), arXiv:1509.07213, abstract read in session). A tilted six-beam MOT loaded from a 2D MOT can therefore supply the atoms.

Grating MOT. A single-beam grating MOT fits geometrically: a 22 mm chip 8 mm below the atoms clears every ring beam by ≥ 6.9 mm, its three 45 deg diffracted beams clear all mirrors and the objective by 6.7 mm [VALUE?: diffraction angle, set by the grating period], and the vertical axis below it is free for 96 mm (to G2) for a retro-reflector. Its single input beam must come from above, along the objective axis: through the objective, injected as a focus in its back focal plane, or through a dichroic before it; the MOT light (852 nm) is the imaging wavelength. However, demonstrated grating MOTs are small for this purpose: 2.46(7)e8 Rb atoms when loaded from a 2D grating MOT (Imhof et al., Phys. Rev. A 96, 033636 (2017), arXiv:1703.07926, abstract read in session), and for Cs only 7.0×10^6 atoms, and only with an additional retro-reflected beam of 69 % of the input intensity; at 24 % no atoms were trapped (Takamizawa et al., Appl. Phys. Lett. 126, 244006 (2025), arXiv:2412.11502, abstract read in session). A Cs grating MOT is therefore one to two orders of magnitude short of a single-shot load of 10^6 sites, and would need iterative accumulation, as in the Atom Computing scheme (Norcia et al.), with stored atoms shielded from the MOT light. We therefore recommend the tilted six-beam MOT, with the grating MOT as a compact fallback for smaller arrays or with iterative loading.

6c. Noise and heating requirements

Model (`scripts/cavity_noise.py`, `configs/cavity_noise.yaml`; `results/cavity_noise/noise.json`). We use the parametric-heating rates of Savard, O'Hara and Thomas, Phys.

Rev. A 56, R1095 (1997) (text read in session): intensity noise gives an energy e-folding time $1/T_I = \pi^2\ \nu^2\ S_\epsilon(2\ \nu)$, and position noise an energy-doubling time $1/T_x = \pi^2\ \nu^2\ S_x(\nu)$ / <x^2>. Here, ν is the trap frequency in Hz, S_ϵ and S_x are one-sided power spectral densities of the fractional intensity and of the trap position, and <x^2> = `k_B` T / (m $(2\ \pi\ \nu)^2$). The script reproduces their numerical examples (3.2×10^{-6} /$\sqrt{\text{Hz}}$ and 3.2×10^{-12} m/$\sqrt{\text{Hz}}$ for 100 s at 10 kHz; `tests/test_cavity.py`). The position term is a constant heating rate, dE/dt = 4 π^4 m $\nu^4\ S_x$, independent of temperature; we quote it for atoms at T = 10 μK [VALUE?: temperature after cooling in the lattice], so that a 10 s doubling time is 1 μK/s, i.e. about one vibrational quantum per second at 20 kHz. The couplings of the self-intersecting ring (DERIVED) follow from the site shift a dϕ / $(2\ \pi)$, with dϕ the change of the phase accumulated between the two passes through the crossing, over the path L1 = L_{rt} / 2: (i) a normal displacement d of one ring mirror at incidence AOI, with the laser locked to the round trip, shifts the sites by a (2 d cos AOI / λ)(1 - L1 / L_{rt}), i.e. by 3.2 d for the in-plane rings and 6.4 d for the vertical ring at AOI = 30 deg; (ii) a laser frequency change dν that the lock does not remove shifts them by a L1 dν / c (7.7×10^{-15} and 2.2×10^{-14} m/Hz); (iii) a change dθ of the crossing half-angle changes the pitch by a cot(θ) dθ and moves the edge sites of the 4 mm array by (S/2) cot(θ) dθ (14.7 mm/rad, in-plane). For the vertical lattice the array is perpendicular to the lattice vector, so a symmetric angle change only rescales the period about the loaded plane at the crossing; a single-arm change dθ tilts the planes by dθ / 2 and moves the edge sites at the half-diagonal by S / (2 sqrt 2) dθ (1.41 mm/rad, Provisional); (iv) a residual laser-cavity detuning dν_r, with linewidth κ (FWHM), modulates the circulating intensity by -(2 dν_r / κ)^2. For Gaussian dν_r with a flat one-sided spectrum S_ν over a bandwidth B = 100 kHz this gives $S_\epsilon \sim 32\ S_\nu^2$ B / κ^4, and with a static lock offset δ_0 the linear term $S_\epsilon = 64\ \delta_0^2\ S_\nu$ / κ^4. The linewidths are those of the over-coupled design point (T = 3L, Section 3) with the 258 MHz (in-plane) and 181 MHz (vertical) free spectral ranges.

Requirements (one-sided amplitude spectral densities; position, displacement, frequency and angle at ν, intensity at 2 ν; lock residual flat over 100 kHz, zero offset):

Lattice, depth	ν	Heating time	RIN at 2 ν	Site position	One mirror	Laser frequency	Crossing angle (edge sites)	Lock residual G = 300 / 1500
In-plane, 0.2 mK	19.8 kHz	10 s	-106 dBc/Hz	1.0 pm/$\sqrt{\text{Hz}}$	0.32 pm/$\sqrt{\text{Hz}}$	132 Hz/$\sqrt{\text{Hz}}$	0.070 nrad/$\sqrt{\text{Hz}}$	22 / 4.4 Hz/$\sqrt{\text{Hz}}$ (7.0 / 1.4 kHz rms)
In-plane, 0.2 mK	19.8 kHz	100 s	-116 dBc/Hz	0.32 pm/$\sqrt{\text{Hz}}$	0.10 pm/$\sqrt{\text{Hz}}$	42 Hz/$\sqrt{\text{Hz}}$	0.022 nrad/$\sqrt{\text{Hz}}$	12 / 2.5 Hz/$\sqrt{\text{Hz}}$
In-plane, 0.86 mK	41.0 kHz	10 s	-112 dBc/Hz	0.24 pm/$\sqrt{\text{Hz}}$	0.074 pm/$\sqrt{\text{Hz}}$	31 Hz/$\sqrt{\text{Hz}}$	0.016 nrad/$\sqrt{\text{Hz}}$	15 / 3.0 Hz/$\sqrt{\text{Hz}}$
Vertical, 0.2 mK	9.9 kHz	10 s	-100 dBc/Hz	4.1 pm/$\sqrt{\text{Hz}}$	0.64 pm/$\sqrt{\text{Hz}}$	186 Hz/$\sqrt{\text{Hz}}$	2.9 nrad/$\sqrt{\text{Hz}}$	22 / 4.3 Hz/$\sqrt{\text{Hz}}$
Vertical, 0.86 mK	20.5 kHz	10 s	-106 dBc/Hz	0.95 pm/$\sqrt{\text{Hz}}$	0.15 pm/$\sqrt{\text{Hz}}$	43 Hz/$\sqrt{\text{Hz}}$	0.68 nrad/$\sqrt{\text{Hz}}$	15 / 3.0 Hz/$\sqrt{\text{Hz}}$

What the table shows. (i) Intensity noise. The cavity passes the laser's relative intensity noise unchanged below κ / 2 (206 kHz at G = 300), so the input beam itself needs -106 dBc/Hz at 40 kHz for 10 s at 0.2 mK, and -116 dBc/Hz for 100 s. This is a requirement on the amplifier and the intensity servo, not on the cavity [VALUE?: RIN of a stabilised 1075 nm fibre amplifier at 20 to 80 kHz]. (ii) Lock quality. The residual detuning must stay below 22 Hz/$\sqrt{\text{Hz}}$ over 100 kHz (7.0 kHz rms, 1.7 % of the 413 kHz linewidth) at G = 300, and below 4.4 Hz/$\sqrt{\text{Hz}}$ (1.4 kHz rms) at G = 1500. The requirement scales as κ^2 in the quadratic regime, so G = 1500 is five times harder. A static offset of 1 % of the linewidth makes the linear conversion comparable (26 and 5.2 Hz/$\sqrt{\text{Hz}}$). This quantifies the reason given in Section 3 for working at G $\sim$ 300,

and is consistent with the frequency-noise heating that Norcia et al. identify in their cavity (190 ms). (iii) Position noise is the stricter condition. A site motion of 1 pm/$\sqrt{\text{Hz}}$ at 20 kHz, i.e. one quantum per second, corresponds to a single-mirror displacement of 0.32 pm/$\sqrt{\text{Hz}}$, or 0.1 pm/$\sqrt{\text{Hz}}$ for 100 s. Coating Brownian noise is far below this `[VALUE?: approx. 1e-17 m/rtHz at kHz, verify]`. The relevant source is the excitation of mount and spacer resonances in the 10 to 40 kHz band by acoustic noise, pumps and the lock actuator. Hence the ring should be a monolithic, low-expansion assembly (Section 7, as in Park et al.), with its lowest mechanical resonances either above $\sim$ 50 kHz or well damped `[VALUE?: mount vibration spectrum]`. (iv) Laser frequency noise. With the laser locked to the ring, its frequency follows the round-trip length, and entry (i) already includes this through the factor (1 - L1 / L_{rt}). The frequency entry of the table applies to laser noise that the lock does not remove, i.e. above the lock bandwidth and below κ / 2; a lock bandwidth above 2 ν avoids it. (v) Crossing angle. Edge sites move by 14.7 mm per rad of crossing-angle change, so 0.07 nrad/$\sqrt{\text{Hz}}$ is needed at 20 kHz. For a mirror mount with a 10 mm lever this is 0.7 pm of differential motion (DERIVED), comparable to (iii), not a separate technology. For the vertical lattice the corrected lever (1.41 mm/rad) relaxes this to 2.9 nrad/$\sqrt{\text{Hz}}$ at 10 kHz (Provisional). (vi) The cavity is a mode filter for input pointing jitter. The nearest transverse mode of the 70 mm ring lies 0.216 rad from the fundamental per round trip, i.e. 8.9 MHz, against a 206 kHz half-linewidth at G = 300. Input misalignment therefore couples into it with $\sim$ 0.02 of the amplitude it would have without the cavity (DERIVED, ratio of half-linewidth to offset), and the site positions are set by the mirrors, not by the input beam. (vii) The vertical lattice, with its lower trap frequency at 0.2 mK, is the least sensitive in intensity (-100 dBc/Hz at 20 kHz), but its larger lever (8 µm period) gives a similar mirror requirement (0.64 pm/$\sqrt{\text{Hz}}$).

These are requirements, not predictions: we have not simulated the time-dependent depth and site position, nor estimated the vibration spectrum of a real mount.

6d. How large a step is 10^6 sites? The same analysis at 10^4 and 10^5

Method (`scripts/cavity_scaling.py`, `configs/cavity_scaling.yaml`; `results/cavity_scaling/scaling.json`, scaling.png). We ran the models of Sections 3 to 6c at 10^4 (100 × 100 sites, 0.40 mm side) and 10^5 (316 × 316, 1.26 mm side) with the same pitch, barrier depth (0.2 mK per axis), sheet thickness (w_z = 75 µm), losses and G = 300: Fox-Li designs and spectra (`configs/cavity_foxli_n1e4.yaml`, `_n1e5.yaml`; depth rms converged to 0.3 % of its value between 4096 and 16384 points), λ/1000 figure errors and 1 µrad tilts on every mirror with a perfect-metrology redesign and the atom loop (`configs/cavity_correction_n1e4.yaml`, `_n1e5.yaml`; correlation lengths scaled with the array side, so the relative error structure is that of the 0.5 and 2 mm cases at 10^6), the vertical ring (`configs/cavity_vertical_plain_n1e4.yaml`, `_n1e5.yaml`) and its mirror intensities (`configs/cavity_vertical_mirrors_plain_n1e4.yaml`, `_n1e5.yaml`). At each size we also ran the 10^6 design unchanged.

The 10^6 design does not scale down. Its 0.2× telescope exists to break the degeneracy of a 5.5 mm flat mode (Fresnel number 53 in the plain ring). A smaller mode is already non-degenerate, and the telescope then compresses it towards a focus on G. At 10^4 the telescope ring has an ideal depth non-uniformity of 4.4 % at 0.73 of the enhancement and a graded-mirror phase of 1262 rad peak to valley; λ/1000 errors give 9.9 and 23 %, and the atom loop fails (28.6 %, G/G0 0.09). At 10^5 the ideal is 0.31 %, and errors give 6.6 and 8.4 %. The vertical ring's 0.15× telescope fails in the same way (ideal 9.8 % at 10^5, `results/cavity_vertical_n1e5`). Hence the adopted design at 10^4 and 10^5 is the plain ring: plane folds, graded mirrors 150 mm from the crossing, no telescope, in-plane and vertical. Its mirrors must be larger than the 10^6 strips in the direction of the small flat-tops, which diffract: with the 10^6 apertures (2.5 mm horizontal, 5 mm vertical half-widths) the 10^5 vertical ring's design target loses 1.6×10^{-3} and 2.0×10^{-2} per round trip (enhancement 0.78); with 8 and 15 mm it loses 1.6×10^{-8} and

3.4×10^{-5} (`results/cavity_vertical_apertures/apertures.json`).

	10^4 sites	10^5 sites	10^6 sites
Array side; in-plane flat-top equivalent width	0.40 mm; 0.55 mm	1.26 mm; 1.74 mm	4.00 mm; 5.51 mm
Hologram tweezers, input	14 W	140 W	1399 W
Lattice in free space, input	69 W	150 W	1041 W
Cavity lattice, input at G = 300 / 1500	0.23 / 0.046 W	0.50 / 0.10 W	3.5 / 0.69 W
Tweezer power / cavity input at G = 300	61	279	403
Vertical ring share of the circulating power	69 %	55 %	79 %
Adopted vertical ring	plain, Gaussian vertical profile (w = 266 μm)	plain, order-10 vertical flat-top	0.15× horizontal, 4× vertical telescope
Adopted in-plane ring	plain, p = 150 mm	plain, p = 150 mm	0.2× telescope
Fresnel number of the plain ring; its nearest mode (phase, loss)	0.52; 0.78 rad, 2.4×10^{-2}	5.3; 0.17 rad, 2.5×10^{-4}	53; 0.017 rad, 2×10^{-9}
Ideal depth non-uniformity, in-plane	0.059 %	0.053 %	0.23 %
$\lambda/1000$ on every mirror: uncorrected; perfect redesign; atom loop	2.0, 3.0 %; 1.3, 0.77 %; 3.0, 2.8 %	1.6, 2.6 %; 0.16, 0.17 %; 0.20, 0.49 %	2.0, 4.0 %; 0.41, 0.49 %; 0.42, 0.75 %
The same, 0.5 mm-scale errors (scaled)	4.2 %; 2.4 %; 5.8 %	6.6 %; 2.2 %; 2.0 %	3.6 %; 2.4 %; 2.3 % (50 mm ring)
1 μrad tilts on every mirror, uncorrected	0.08, 0.10 %	1.1, 0.19 %	1.1 %
The plain ring at this size, $\lambda/1000$, uncorrected	2.0, 3.0 %	1.6, 2.6 %	20, 42 %
Peak intensity on the hottest in-plane mirror	4.2 kW/cm^2 (fold)	4.2 kW/cm^2 (fold)	48 kW/cm^2 (P2)
Largest absorbing defect, in-plane, fused silica / sapphire	1.6 / 39 μm	1.6 / 39 μm	0.14 / 3.4 μm
Vertical ring: ideal; $\lambda/1000$ uncorrected (h, v); after atom loop	0.24 %; 2.9, 3.4 % (h), 0.28, 0.24 % (v); 0.43 % (one run)	0.20 %; 10, 12 % (h), 1.6, 4.1 % (v); 1.4 % (one run)	0.073 %; 5.7, 12 % (h), 5.4, 7.3 % (v); 1.06, 0.96 % (best round)
Vertical ring, hottest mirror	25 kW/cm^2 (P1)	7.4 kW/cm^2 (P1)	55 kW/cm^2 (G, P2')
Edge-site crossing-angle requirement, 10 s at 10 μK	0.70 nrad/$\sqrt{\text{Hz}}$	0.22 nrad/$\sqrt{\text{Hz}}$	0.070 nrad/$\sqrt{\text{Hz}}$
MOT atoms needed (cloud matched at the corners; into the sheets)	4.7×10^6 to 5.0×10^6; 3.7×10^5	1.5×10^8 to 1.6×10^8; 3.7×10^6	4.8×10^9 to 5.1×10^9; 3.7×10^7

What the comparison shows. (i) Power. Laser power is not a constraint at 10^4 (14 W of tweezers), and a lattice in free space needs more than tweezers below about 10^5, because its in-plane power scales as the array side and the tweezer power as the number of sites. The cavity reduces the input 61, 279 and 403 times against tweezers at 10^4, 10^5 and 10^6. Hence the cavity is a convenience at 10^4, a large saving at 10^5 (0.50 W against 140 W) and necessary at 10^6. The vertical ring takes 55 % of the power at 10^5 and 79 % at 10^6, because its vertical flat-top must cover a walk-off that grows with the array side; at 10^4 its share is 69 % because the adopted Gaussian profile (item (iv)) is wider than a flat-top would be. A sheet thickness w_z scaled as sqrt(side), which keeps the same 1 % roll-off per array length, would lower the 10^4 input to 0.18 W (with the adopted Gaussian vertical ring) and the 10^5 input to 0.40 W (DERIVED; w_z = 24 and 42 μm). (ii) Mode control. The plain ring is non-degenerate at 10^4 and 10^5 (nearest

mode 0.78 and 0.17 rad away with 2.4×10^{-2} and 2.5×10^{-4} loss), so no telescope is needed. The same $\lambda/1000$ errors that give 20 to 42 % in the plain 10^6 ring give 1.6 to 3.0 % at 10^4 and 10^5, and tilts become nearly harmless (0.08 to 0.10 % at 10^4). The atom loop corrects 10^5 to 0.20 to 0.49 %, but not 10^4 (2.8 to 3.0 %), where a perfect-metrology redesign still reaches 0.77 to 1.3 %. A follow-up run (`configs/cavity_correction_n1e4_loop.yaml`) shows two causes. First, the 48-bin fit over 100 sites per side overfits: with 24 bins the loop reaches 1.1 and 2.2 % at its best round (the first or second). Second, the phase-only floor is itself 0.77 to 1.3 % at this size, and the loop drifts above it after its best round (to 1.9 and 4.2 % after six). A camera loop on the transmitted beams (5 µm pixels, 24 bins) behaves the same way: best 1.3 and 1.9 % after two to three rounds, then drifting to 2.1 and 2.2 % after six. Hence correction at 10^4 is limited to about 1 % by the phase-only graded mirrors, and the loop needs a stopping rule. (iii) Defects and heat. The intensity at the atoms is set by the depth, 19.8 kW/cm^2 per beam at every size. Without the telescope the in-plane mirrors see at most 4.2 kW/cm^2, so the defect limit is 1.6 µm on fused silica (39 µm on sapphire) at 10^4 and 10^5, against 0.14 µm at 10^6 (P2 of the telescope ring, 48 kW/cm^2). The 10^5 vertical ring needs no vertical telescope: its hottest mirror sees 7.4 kW/cm^2, against 170 kW/cm^2 for the 10^6 ring as folded. (iv) Vertical ring at 10^4. Its vertical flat-top is only 19 µm high (0.057 mm equivalent height) and diffracts strongly over the 414 mm half trip: even with ±25 mm mirror height in the ring plane the ideal non-uniformity is 3.2 % and $\lambda/1000$ errors reduce the enhancement to as little as 0.10. A Gaussian vertical profile is the better choice at this size, and we adopt it: a 266 µm waist keeps the intensity within 1 % over the ±19 µm walk-off and has a 207 mm Rayleigh range. Simulated (`configs/cavity_vertical_gauss_n1e4.yaml`), it gives 0.24 % ideal non-uniformity, vertical-direction $\lambda/1000$ errors are nearly harmless (0.24 and 0.28 %), horizontal ones give 2.9 and 3.4 %, and atom feedback reaches 0.43 % (one run, enhancement 0.86 / 0.97). The price is 5.9 times the vertical power, 45 W circulating, and 0.23 W of total input at G = 300; its hottest mirror (P1) sees 25 kW/cm^2, the intensity at the atoms scaled by the beam size on P1. (v) Noise. Only the crossing-angle requirement depends on size (0.70, 0.22 and 0.070 nrad/$\sqrt{\text{Hz}}$); the intensity, lock and mirror-displacement requirements of Section 6c are the same at every size, because the trap frequency is. (vi) Loading. A MOT of about 5×10^6 atoms suffices for a single-shot load at 10^4, which the demonstrated Cs grating MOT reaches (7.0×10^6 atoms, with an added retro-reflected beam, Section 6b); at 10^5 a grating MOT suffices only if the atoms are accumulated in the sheets (3.7×10^6 needed there), and at 10^6 it does not.

Hence the step from 10^5 to 10^6 is the large one. Up to 10^5 the design is a plain ring with mirrors of modest quality: 0.5 W of input, $\lambda/1000$ errors at the per-cent level that the atom loop corrects, micron-scale defect tolerance, no telescope and, at 10^4, a grating MOT. At 10^6 the mode becomes degenerate, which brings the telescope, the 0.14 µm defect limit, a vertical ring that needs its own vertical telescope and sapphire mirrors (55 kW/cm^2), the 0.07 nrad/$\sqrt{\text{Hz}}$ angle stability and a 10^9-atom MOT.

7. What must improve, and next steps

What must improve. (i) Figure and alignment of the large telescope mirrors: $\lambda/1000$ rms figure and < 1 µrad tilt for ≤ 4 % depth non-uniformity before correction. Static errors of this size are corrected to ~ 0.5 % by closed-loop re-shaping of the graded mirrors (Section 4b), so the strict tolerance applies to drift between corrections. Drift at the µrad level calls for a monolithic, low-expansion assembly as in Park et al. (ii) Cleanliness of every surface near the waist at the level of point absorbers below 0.14 µm on fused silica (P2), or sapphire substrates (23× more tolerant). (iii) Losses of the graded mirrors $\leq 1 \times 10^{-3}$ per bounce, i.e. figured surfaces rather than metasurfaces. (iv) The vertical lattice, which dominates the power at 10^6 sites (785 of 989 W circulating); the adopted 4× vertical telescope keeps its mirrors at ≤ 55 kW/cm^2 on sapphire, at about 1 % depth non-uniformity after feedback (Section 5). (v) Single-plane loading in an 8

μm vertical lattice from a MOT of 10^8 to 10^9 atoms, and site-resolved rearrangement into the lattice; neither is simulated here (Section 6b gives the atom budget).

Next simulations. (i) The vertical ring with propagation along an array side (a narrower beam) as an alternative to its diagonal layout. (ii) A time-dependent simulation of depth and site position under laser and lock noise, and a vibration model of the ring mounts, against the requirements of Section 6c. (iii) The vertical ring's thermal bulge (0.12 nm, Section 5b) fed into Fox-Li (done for the in-plane rings, Section 5). (iv) A model-based closed loop for large starting errors (a lower gain was tested and does not help, Section 4b), and the effect of drift during a correction cycle. (v) Mounts, MOT coils and the vacuum envelope around the layout of Figures 3 and 5, and the windows for the tilted MOT and pumping beams. (vi) Loading dynamics: accumulation of atoms in the in-plane sheets from the tilted MOT and plane selection. (vii) A stopping rule or a model-based update for the vertical-ring loop, which drifts after its best round (Section 4b).

Verification notes

Provenance ledger:

Value	Location	Type	Pointer
Power budget, losses, enhancements	Result, Sections 1, 3	DERIVED (code)	`results/cavity_budget/budget.json`
Tweezer anchors 143 and 80 mK x site / W	Section 3	SIM (earlier runs)	`docs/scaling_100k.md`, `docs/million_sites.md`
Cs depth per intensity 2.53×10^{-13} K / (W/m^2)	Section 3	DERIVED from SOURCE	`atoms/alkali.py` (Steck data)
Concentric ray trace	Section 1	SIM, review before submission	`results/cavity_tweezer_bounds/bounds.json`
Fox-Li designs, spectra, tolerances	Sections 4, Result	SIM, review before submission	`results/cavity_foxli/foxli.json` (field .npz files are regenerated by the script, not versioned)
Exploratory curved-fold spectra	Section 4	SIM (exploratory, not saved)	session scratch scripts
Correction study (open and closed loop, strokes)	Section 4b, Result	SIM, review before submission	`results/cavity_correction/correction.json`
Camera-loop correction	Section 4b, Result	SIM, review before submission	`results/cavity_correction/correction.json` (keys `*_closedloop_camera_*`)
Transmitted power ~ 5 mW for 50 ppm at 100 W	Section 4b	DERIVED	5×10^{-5} x 100 W
2D separability check	Section 4b	SIM, review before submission	`results/cavity_2d_check/check2d.json`
Vertical ring uniformity and plane flatness	Section 4b	SIM, review before submission	`results/cavity_vertical/vertical.json`
Vertical ring at 0.1×, vertical flat-top edge orders	Section 4b, Section 2	SIM (exploratory, not saved)	session scripts
Advanced LIGO substrate figure 0.08 to 0.23 nm rms (spec $<$ 0.3 nm, 160 mm); coated 0.18 to 0.69 / 0.15 to 0.31 nm	Section 4b	SOURCE	arXiv:1411.4547, Table 4 and Section 4 text (PDF text read in session)
LIGO metrology interferometer repeatability $<$ 0.15 nm rms required, 0.1 nm goal	Section 4b	SOURCE	LIGO-E080508-v2 (PDF text read in session)

Value	Location	Type	Pointer
LIGO ETM polishing spec < 0.3 nm rms over 160 mm, piston/tilt/power removed	Section 4b (via arXiv table)	SOURCE	LIGO-E080512-v5 (PDF text read in session)
Zeiss EUV metrology 27 pm repeatability, 46 pm reproducibility; M6 figure 0.10 nm rms	Section 4b	SOURCE	Kurz et al., EUVL Symposium 2004, `https://euvlsymposium.lbl.gov/pdf/2004/presentations/day3/To03_Kurz.pdf` (PDF text read in session)
70 mm ring: design, spectrum, correction check	Section 2, Figure 4	SIM, review before submission	`results/cavity_foxli/foxli.json` (`design_telK_M0.2_d70`), `results/cavity_correction_d70/correction.json`
Two-ring layout clearances, angles, vertical design at 70 mm	Sections 2, 6, Figure 3	DERIVED (code)	`results/cavity_layout/layout.json`
Design sketches	Figures 3, 4	DERIVED / SIM	`scripts/plot_cavity_sketch.py`
Vertical-ring fold geometry and clearances	Section 2, Figure 5	DERIVED (code)	`results/cavity_vertical_layout/vertical_layout.json`
Vertical ring, adopted geometry: uniformity, tolerances, closed loop	Section 4b	SIM, review before submission	`results/cavity_vertical_chosen/vertical.json`
Vertical ring with the in-plane telescope at 60 mm (0.07 rad, 1×10^{-5}; 7.7 to 69 %)	Section 2	SIM (exploratory, not saved)	session scripts
Optical access map, MOT triad, pumping / Raman lines, grating-MOT checks	Section 6b, Result	DERIVED (code)	`results/cavity_access/access.json`, access.png
Atom budget for loading (1.6×10^{10} cm^{-3}, $\sim 5 \times 10^{9}$, $\sim 4 \times 10^{7}$)	Section 6b	DERIVED	site volume, Gaussian cloud, 19 planes in the sheets
Cs molasses 1.1(1)e9 atoms, $\sim 1 \times 10^{10}$ atoms/s source	Section 6b	SOURCE	NIST, `https://www.nist.gov/node/735766` (Donley, Heavner, Jefferts)
Cs 2D-MOT flux 4.02×10^{10} atoms/s	Section 6b	SOURCE	Huang et al., Chin. Phys. B 25, 063701 (2016), arXiv:1509.07213
Rb 3D grating MOT 2.46(7)e8 atoms from a 2D grating MOT	Section 6b	SOURCE	Imhof et al., Phys. Rev. A 96, 033636 (2017), arXiv:1703.07926
Cs grating MOT 7.0×10^{6} atoms with a 69 % retro beam; none at 24 %	Section 6b	SOURCE	Takamizawa et al., Appl. Phys. Lett. 126, 244006 (2025), arXiv:2412.11502
Noise requirements (RIN, site position, mirror, frequency, angle, lock residual)	Section 6c	DERIVED (code)	`results/cavity_noise/noise.json`
Heating rates $1/T_I = \pi^2\ \nu^2\ S_\epsilon(2\ \nu)$, $1/T_x = \pi^2\ \nu^2\ S_x$ / $\langle x^2 \rangle$ and their numerical examples	Section 6c	SOURCE	Savard, O'Hara, Thomas, Phys. Rev. A 56, R1095 (1997) (text read in session)
Noise couplings of the self-intersecting ring, lock-residual conversion, mode-filter ratio ~ 0.02	Section 6c	DERIVED	`scripts/cavity_noise.py` docstring; 206 kHz / 8.9 MHz

Value	Location	Type	Pointer
Lower-gain closed loop (gain 0.25, 0.1)	Section 4b	SIM, review before submission	`results/cavity_correction/correction.json` (keys `*_gain*`)
Absorption bulge fed into Fox-Li	Section 5	SIM, review before submission	`results/cavity_thermal_foxli/thermal_foxli.json`
Vertical-ring mirror intensities, defect limits, z-telescope variants	Section 5, Result, Design summary	SIM, review before submission	`results/cavity_vertical_mirrors/mirrors.json`
Input power per ring, design summary	Design summary	DERIVED	P_{circ} / (0.95 G), `results/cavity_budget/budget.json`
Scaling study at 10^4 and 10^5: designs, spectra, errors, correction, vertical ring, mirror intensities	Section 6d, Result (viii)	SIM, review before submission	`results/cavity_foxli_n1e4`, `_n1e5`; `results/cavity_correction_n1e4`, `_n1e5`; `results/cavity_vertical_plain_n1e4`, `_n1e5`; `results/cavity_vertical_mirrors_plain_n1e4`, `_n1e5`; `results/cavity_vertical_n1e5`
Scaling comparison: powers, defect limits, angle requirement, MOT atom numbers	Section 6d	DERIVED (code)	`results/cavity_scaling/scaling.json`
Gaussian vertical profile at 10^4 (266 µm, 5.9×, 0.23 W)	Section 6d	DERIVED	w = `h_z` / sqrt(-ln 0.99 / 2); sqrt(π/2) w / H_{eq}
Vertical ring with 4× / 8× vertical telescope: errors, closed loop (6 and 12 rounds)	Sections 4b, 5, Design summary	SIM, review before submission	`results/cavity_vertical_ztel4`, `_ztel8`, `_ztel4_r12` /vertical.json
Vertical-ring strip heights: beam extents, layout check	Sections 2, 5	SIM / DERIVED (code)	`results/cavity_vertical_beam_heights/heights.json`, `results/cavity_vertical_layout_check/check.json`
10^4 loop diagnosis (24 bins, camera); Gaussian vertical ring at 10^4	Section 6d	SIM, review before submission	`results/cavity_correction_n1e4_loop`, `results/cavity_vertical_gauss_n1e4`, `results/cavity_vertical_mirrors_gauss_n1e4` (intensities scaled by the power ratio 5.87 in `scripts/cavity_scaling.py`)
Heat budget: power flow, absorbed and scattered power, mirror temperatures and bulge, time constants, stability table	Section 5b, Result (iv)	DERIVED (code) / SIM (conduction model)	`results/cavity_heat_budget/heat.json`
Vertical-mode spectra converged on grids of 2048 to 16384 points (no telescope, 4×, 8×, horizontal)	Sections 4b, 5	SIM, review before submission	`results/cavity_vertical_spectra_check/spectra.json`
Strip-aperture design losses of the vertical ring (10^6) and the 10^5 / 10^4 rings with 10^6 apertures	Sections 5, 6d	SIM, review before submission	`results/cavity_vertical_apertures/apertures.json`

Value	Location	Type	Pointer
50 mm ring vertical design (f_z 30 mm, R 166 mm, m 0.07, mode spacing 3.0 rad, 0.24 / 0.26 / 0.65 mm) and the 70 mm design with P2 size	Section 2	DERIVED (code)	`results/cavity_layout/layout.json` (`reference_vertical_designs`, `vertical_designs`)
Unconstrained vertical ring's horizontal spectrum (0.14 rad, 3.1×10^{-4})	Section 4b	SIM	`results/cavity_vertical/vertical.json` (`spectrum_x`, rerun with `sparse_spectrum`)
Lock correction	Section 4	SIM	104 strongly perturbed cases were rerun with the self-consistent lock (25 had locked off the fundamental); their earlier values are in `results/cavity_foxli/relock_before.json`
SiN phase slope vs width	Section 4	DATA (analysis of library)	`data/library_main.h5`, `data/library_est_1075.h5`
Vertical Gaussian design, layout	Section 2, 6	DERIVED (code)	`results/cavity_layout/layout.json`
Intensities, defects, deformation	Section 5	SIM / DERIVED, review before submission	`results/cavity_thermal/thermal.json`
Trap frequencies, registration	Section 6	DERIVED (code)	`results/cavity_lattice/lattice.json`
Gaussian waist 32 mm and 7× power for 1 % flatness	Result	DERIVED	w = h sqrt(2 / -ln 0.99); ratio sqrt(π/2) w / W_{eq}
Sheet-only vertical frequency 0.7 kHz	Section 2	DERIVED	sqrt(4 × 2U / (m w_z^2))
Park et al.: F 5025, G 1132, 489 µm, 80 mW, ∼ 92 W, ∼ 100 ppm, w^4 sensitivity, 1 % fringes	Section 2	SOURCE	arXiv:2110.08073 (PDF text read in session)
Norcia et al.: F 2900 / 3000, waists 268 / 183 µm, ≤ 50 ppm per mirror, 1225 sites, ∼ 100×, 190 ms heating	Sections 2, 3	SOURCE	arXiv:2401.16177v3 (HTML read in session)
Belanger and Pare graded-phase resonator	Section 2	SOURCE (abstract)	DOI 10.1364/OL.16.001057
LIGO point absorbers	Section 5	SOURCE (abstract)	DOI 10.1364/AO.419689, arXiv:2101.05828

Open placeholders: `[VALUE?: AR loss per surface 2.5e-4 to 1e-3]`, `[VALUE?: Brewster residual loss 1e-4]`, `[VALUE?: IBS absorption 1 ppm]`, `[VALUE?: refractive objective 0.95, ideal relay 0.995 per pass]`, `[VALUE?: mode-converter coupling 0.95]`, `[CONFIRM?: cell wall 15 mm from the atoms; cell size for the layout]`, `[REF?: freeform figuring at nm rms over mm apertures]`, `[VALUE?: absolute flat-calibration agreement approx. 0.4 nm]`, `[VALUE?: loss and stability of an intracavity deformable mirror]`, `[CONFIRM?: objective position and mount, chamber windows, mounts]`, `[VALUE?: grating diffraction angle]`, `[VALUE?: ~0.5 collisional filling]`, `[VALUE?: atom temperature 10 uK in the lattice]`, `[VALUE?: RIN of a stabilised 1075 nm fibre amplifier]`, `[VALUE?: coating Brownian noise approx. 1e-17 m/rtHz]`, `[VALUE?: mount vibration spectrum at 10 to 40 kHz]`, `[VALUE?: tolerable vertical depth non-uniformity]`, `[VALUE?: expansion coefficients of ULE, Invar, stainless steel, aluminium]`, `[VALUE?: emissivity 0.9]`, `[VALUE?: absorption of`

a figured graded mirror], [VALUE?: vibration of a water-cooled mount], [VALUE?: MOT coil dissipation], [VALUE?: amplifier cooling]. Unverified: Nshii et al., Nature Nanotechnology 8, 321 (2013), DOI 10.1038/nnano.2013.47, grating-MOT atom number (seen only in a search snippet, not used).

Sourcing gaps. The 12,000-finesse microstructured-mirror cavity (Dickmann et al., Commun. Phys. 2023, DOI 10.1038/s42005-023-01131-1) and the Si3N4 photonic-crystal membrane mirror (arXiv:2401.16695, R $\sim$ 99 %) were seen only in search snippets and are not used for any number. The Weiss group's large-spacing Cs lattice (Nelson, Li and Weiss, Nat. Phys., DOI 10.1038/nphys645) is a relevant precedent for imaging single atoms at $\sim$ 5 µm spacing; its spacing (4.5 or 5 µm) is unverified.